

CompTIA A+ Core 1 (220-1201)

1.0 Mobile Device

▼ 1.1 Laptop hardware

▼ Laptop battery

1. **The power power source when unplugged**
 - May be easily replaced or require a teardown
 - Five minutes vs one hour
2. **Lithium-Ion (Li-ion) and Lithium-Ion polymer (Lipo) are common**
 - No “memory effect”
 - Charging the battery does not diminish capacity
3. **Different form factor for each laptop**
 - Battery types and styles can vary
 - Battery technology is changing constantly

▼ Laptop keyboard

1. The most used component of the laptop
 - Can be easy to replace
 - A few screws and a single ribbon cable
 - May not always be this simple
2. Or connect an external USB keyboard
 - not very portable but works in a pinch
3. Some repairs might require the removal or replacement of a key cap
 - This can be a delicate procedure

▼ Laptop Memory

1. Small Outline Dual In-line Memory Module (SO-DIMM)
2. Memory used in laptops and mobile devices
 - Often easy to install and replace

3. Some laptop memory is soldered to the system board

- No upgrade available
- Requires a full system board replacement

▼ Laptop Storage

1. Magnetic disk

- Traditional spinning drive platters
- 2.5 inch form factors (3.5 inch for desktops)

2. SSD (Solid- State drive)

- All memory, no moving parts
- Silent, fast access time, less latency
- 2.5 inch form factors

3. M.2

- Smaller form factor
- No SATA data or power cables
- Easy to install and replace

▼ Wireless and Bluetooth

1. Wireless network connectivity

- Often integrated into the motherboard
-

2. 802.11

- Local area network (LAN)
- High speed
- Internet access

3. Bluetooth

- Personal area network (PAN)
- Short range
- Connect peripherals and other nearby devices

▼ Biometrics

1. Sign in or unlock your laptop with a fingerprint reader or face recognition

- Something you are

-

2. Requires additional configuration in the OS

- Hardware required for most options

-

3. Relatively secure

- Faces and fingerprints are quite unique

-

▼ Near-field communication (NFC)

1. Common on mobile phones and smart watches

- Payment method on your wrist

-

2. Short-distance networking

- 4 centimeters or less
- Data transfers or authentication

3. Use it for authentication without typing a password

- Hospital workstations
- Warehouses
- Manufacturing

▼ Wi-Fi antennas

1. Multiple antennas

- WiFi main and aux
- Bluetooth

2. Antenna wires wrap around the laptop screen

- It's up high !

- The actual antenna "foils" are usually taped to the top or sides of the screen.

▼ Camera

1. Video capture
 - Built-in camera and microphone
 -
2. Usually include both audio and video
 - Specialized drivers and software
 -

▼ Microphone

1. Build-in to the laptop display
 - Useful for video calls
 -
2. Not good for non-casual use
 - Analog or USB microphones can help
 -

▼ 1.2 Connecting Mobile Devices

▼ Universal Serial Bus (USB)

1. USB
 - high-speed wired communication
 - Used for charging and data transfers
2. Micro- USB
 - a smaller USB connection
 - Common now worldwide
3. Older devices may use Mini- USB
 - Slightly larger

▼ USB-C

1. 24-pin double -sided USB connector

- Used for both hosts and devices
-
- 2. Acts as a USB 2.0/3.0/3.1/4 connection
 - Different signals can use the same connector
 -
- 3. Can transmit other signals
 - DisplayPort
 - HDMI
 - Thunderbolt

▼ Lightning

1. Apple proprietary
 - 8-pin digital signals
 - iPhone and iPad devices
2. Some advantages over Micro-USB
 - Higher power output for phones and tablets
 - can be inserted either way
 - Simpler design

▼ NFC

1. Near Field Communication
 - Send small amounts of data wirelessly over a limited area
 -
2. Built into your phone
 - Payment systems, transportation, in-person information exchange
 -
3. Access token, identity "card"
 - Short range with encryption support
 - Device pairing and data exchange

▼ Bluetooth

1. High speed communication
 - PAN -personal area network
 -
2. Connects our mobile phone
 - Smartphones
 - Tethering
 - Headsets and headphones
 - Health monitors
 - Automobile and phone integration
 - Smartwatches
 - External Speakers

▼ Hotspot / Tethering

1. Turn your phone into a WIFI hotspot
 - Your own personal wireless router
 - Extend the cellular data network to all of your devices
2. Dependent on phone type and provider
 - May require additional charges and data costs
 -

▼ Stylus

1. Digital stylus
 - A more advanced writing tool
 -
2. The stylus communicates directly to the device
 - Pressure sensitivity, programmable buttons
 -
3. Must be compatible with the tablet
 - e.g: Apple iPad uses an Apple Pencil

-

▼ Headsets

1. Hands-free audio

- Earphones and microphone
-

2. Wired

- USB connections are common on laptops
- Connect to 3.5 mm TRRS (tip-ring-ring-sleeve) connector
- Analog audio jack
- iPhone can use Lightning port

3. Wireless

- Bluetooth headsets
-

▼ Speakers

1. Mobile

- Battery powered
-

2. Wireless connection

- Bluetooth link
-

3. Stereo sound

- small package
-

▼ Camera /Webcam

1. Internal or external

- Commonly external on desktops, internal on laptops/ tablets/ phones
-

2. Accessible from multimedia applications, photo utilities, and video conferencing

- Real-time video communication
-

▼ Docking station

1. Docking station

- Use external keyboard and mouse
- Extend existing laptop interfaces
- Add additional functionality
- Desktop adapter cards
- Avoid cable issues

▼ Port replicator

1. Similar to a docking station

- Does not commonly have an expansion card option
- Usually connects using USB

▼ Trackpad

1. Replace the mouse

- Useful in tight working areas
- Common on laptops

2. External option

- Battery powered
- Bluetooth connected

3. Drag and tap

- Or use multiple finger input for right-clicking, zooming, and window control
- provides the function of a pointing device

▼ Drawing pad

1. Use an active stylus with an external digitizer

- Very precise input
 -
 - 2. Third-party device
 - Support across many operating systems
 -
- ▼ Track point
1. Track point
 - small pointing stick embedded in a keyboard used for precise cursor control.
 - used as an ergonomic alternative to a mouse or touchpad.

▼ 1.3 Mobile Device Networks

- ▼ Cellular network
1. Mobile device
 - "Cell" Phones
 2. Separate land into "cells"
 - Antenna coverages a cell with certain frequencies
 3. Enable or disable voice and/ or data
 - Airplane mode disables all cellular networks
 - Could also disable data but keep voice
- ▼ 3G technology
1. 3rd Generation
 - Introduced in 1998
 -
 2. Upgraded data connectivity
 - Incremental 3G updates improved speeds
 - Usually several megabits per second
 3. Bandwidth improvement allowed new functionality

- GPS
- Mobile television
- Video on demand
- Video conferencing

▼ 4G and LTE

1. Long Term Evolution (LTE)

- A "4G" technology
- Based on GSM (Global System for Mobile Communications) and EDGE (Enhanced Data Rates for GSM Evolution)
- Standard supports download rates of 150 Mbit/s

2. LTE Advanced (LTE-A)

- Standard supports download rates of 300 Mbit/s

-

3.

-

-

▼ 5G

1. Fifth generation cellular networking

- Launched worldwide in 2020
-

2. Significant performance improvements

- At higher frequencies
- Eventually 10 gigabits per second
- Slower speeds from 100-900 Mbit/s

3. Significant IoT impact

- Bandwidth becomes less of a constraint
- Larger data transfers
- Faster monitoring and notification

- Additional cloud processing

▼ WiFi

1. Connect to a local 802.11
 - Data access for email and Internet
 - Higher speeds, no cellular bandwidth use
2. Requires a local access point
 - Limit distance coverage
 -
3. May also support voice communication
 - WiFi calling
 - Requires support from the voice carrier
 - Very useful when cellular coverage may not be available

▼ Hotspot

1. Turn your phone into a WiFi hotspot
 - Your own personal wireless router
 - Extend the cellular data network to all of your devices
2. Dependent on phone type and provider
 - May require additional charges and data costs
 -

▼ Subscriber Identity Module (SIM)

1. Subscriber Identity Module (SIM)
 - Often tiny physical card
 - Replace the SIM and the phone has a new number
2. Contains information for the carrier and user
 - SIM ID and phone number
 - Cellular network information
 - Storage space for contacts, messages, etc.
3. eSIM - embedded SIM

- SIM is inside the device
 - Cannot be physically removed
4. Many devices support multiple SIM

▼ Bluetooth pairing

1. Connect a Bluetooth device
 - Built-in security
 - Use or verify a PIN
2. Pair the devices one time
 - Future connections should be automatic
3. Enable Bluetooth on both devices
 - Android and iOS: Settings /Bluetooth
4. Set devices to discoverable mode
 - May require key sequence on Bluetooth device
5. Select discovered device
 - Many devices may appear!
6. Enter or confirm PIN
 - Should be the same on both devices
7. Test connectivity
 - Devices should now communicate

▼ GPS

1. Created by the U.S. Department of Defense
 - Over 30 satellites currently in orbit
2. Precise navigation
 - Need to see at least 4 satellites
3. Determines location based on timing differences
 - Longitude, latitude, altitude
4. Mobile device location services and geotracking
 - Maps, directions

- Determine physical location based on GPS, WiFi, and cellular towers

▼ 1.3 Mobile Device Management

▼ Mobile Device Management (MDM)

1. Manage company-owned and user-owned mobile devices
 - BYOD - Bring Your Own Device
 -
2. Centralized management of the mobile devices
 - Specialized functionality
 -
3. Set policies on apps, data, camera, etc
 - Control the remote device
 - The entire device or a "partition"
3. Manage access control
 - Force screenshot and PINs on these single user devices

▼ BYOD

1. Bring your own device
 - Bring you own technology
 -
2. Employee own the device
 - Need to meet the company's requirement
 -
3. Difficult to secure
 - it's both a home device and a work device
 - How is data protected?
 - What happens to the data when a device is sold or traded in?

▼ COPE

1. Corporate owned, personally enabled

- Company buys the device
 - Used as both a corporate device and a personal device
2. Organization keeps full control of the device
 - Similar to compan-owned laptops and desktops
 -
 3. Information is protected using corporate policies
 - Information can be deleted at any time
 -
 3. CYOD - Choose your own device
 - Similar to COPE, but with the user's choice of device
 -

▼ MDM policy enforcement

1. Corporate email configuration
 - User does not need to configure anything
 - The MDM makes the changes on the device
 - Account details, server address, communication method
2. Two factor authentication
 - Require specific authentication types
 - Biometrics, pseudo-random authentication app
3. Corporate applications
 - Allow or restrict app installation
 - Prevent unauthorized app usage

▼ Synchronizing data

1. Many settings are preconfigured
 - Telephone /Text messaging
 -
2. Email
 - Everyone handles email services differently

- Corporate email configurations can vary
- 3. Data synchronization
 - Based on data rates and speeds
 - Important for backup and recovery
- 3. Data types
 - Calendar
 - Contact
- 4. Data caps and transfer costs
 - May have significant financial consequences
 - Cellular/ Mobile vs 802.11
- 2. Verify data restrictions with the wireless carrier
 - Get it in writing
- 3. Enable or disable network connections
 - Control the use of cellular downloads
- ▼ Business applications
 - 1. Mail, cloud storage, etc
 - Many different services available
 - 2. Authenticate
 - Username, password
 - 3. Select the items to synchronize
 - Change will appear on the mobile device
 - 3. Same process for other services
 - Often built-in to mobile device

2.0 Networking

▼ 2.1 Introduce to IP (Internet Protocol)

▼ TCP and UDP

1. Transported inside of IP
 - Encapsulated by the IP protocol
2. Two ways to move data from place to place
 - Different features for different applications
3. OSI Layer 4
 - The transport layer
 -
3. Multiplexing
 - Use many different applications at the same time
 - TCP and UDP

▼ TCP - Transmission Control Protocol

1. Connection-oriented
 - A formal connection setup and close
 - connection-oriented protocol using a three-way handshake
2. Reliable delivery
 - Recovery from errors, guarantee data correct
 - Can manage out of order messages or retransmissions
3. Flow control
 - The receiver can manage how much data is sent

▼ UDP - User Datagram Protocol

1. Connectionless
 - No formal open or close to the connection
2. Unreliable delivery
 - No error recovery
 - No reordering of data or retransmissions

3. No flow control

- Sender determines the amount of data transmitted

▼ Why would you ever use **UDP**

1. Real-time communication

- There's no way to stop and resend the data
- Time doesn't stop for your network

2. Connectionless protocols

- DHCP (Dynamic Host Configuration Protocol)
- TFTP (Trivial File Transfer Protocol)

3. The data might not get through

- The application keep track and decide what to do
- It might not do anything

▼ Communication using **TCP**

1. Connection-oriented protocols prefer a "return receipt"

- HTTPS (Hypertext Transfer Protocol Secure)
- SSH (Secure Shell)

2. The application doesn't worry about out of order frames or missing data

- TCP handles all of the communication overhead
- The application has one job

▼ 2.1 Common Ports

▼ FTP - File transfer protocol (TCP Port 20/21)

1. TCP 20 (active mode data) Data Port / TCP 21 (control) Control Port

- to access, store, and manage shared documents centrally over a network.
- FTP data connection for transferring file data - TCP port 20
-

2. Authenticates with a username and password

- Some system use generic/anonymous login
-
- 3. Full featured functionality
 - List, add, delete, etc
 -
- ▼ SSH - Secure Shell (TCP Port 22)
 1. Encrypted communication link
 2. Use to remote login to computer systems by users
 2. Look and acts the same as Telnet
- ▼ Telnet - Telecommunication network (TCP Port 23)
 1. Login to devices remotely
 - Console access
 2. In the clear communication
 - Not the best choice for production systems
- ▼ SMTP - Simple mail transfer Protocol (TCP Port 25)
 1. SMTP - Simple mail transfer Protocol
 - Server to server transfer
 - TCP port 25
 2. Also use to send mail from device to mail server
 - Commonly configure on mobile device and email
 3. Other protocol are used to client to receive email
 - IMAP
 - POP3
- ▼ DNS - Domain Name System (TCP/UDP Port 53)
 1. Convert name to IP address
 - www.xxx123.com = 162.159.xxx.xxx
 2. There are very critical resources
 - Usually multiple DNS servers are in production

▼ DHCP - Dynamic Host Configuration (UDP Port 67/68)

The server receive responses on port 67 from client requests, while the client receive responses port 68 from server responses

1. Automated configuration of IP address, subnet mask, and etc
2. Required a DHCP server
 - Server, appliance, integrated in to SOHO router, etc
3. Dynamic / pooled
 - IP addresses are assign in real time from a pool
 - Each system is given a lease and must renew at set intervals
3. DHCP reservation
 - Addresses are assigned by MAC address in the DHCP server
 - Manage addresses from one location

▼ HTTP and HTTPS - Hypertext transfer protocol (TCP Port 80) / Hypertext transfer protocol secure (TCP Port 443)

1. Hypertext transfer protocol (TCP Port 80)
 - Communication in the browser
 - And by other applications
2. In the clear or encrypted
 - Support by nearly web servers and clients
3. Hypertext transfer protocol secure (TCP Port 443)
 - Web server communication with encryption

▼ POP3/ IMAP - Post office protocol v3 (TCP Port 110) / Internet Message access protocol (TCP Port 143)

1. Receive emails from an email server
 - Authenticate and transfer
2. Post office protocol v3 (TCP Port 110)
 - Once you "pick it up" from the post office (the server)asic mail transfer functionality

- Cannot see those emails on a second device because they no longer exist on the server
3. Internet Message access protocol (TCP Port 143)
 - IMAP syncs your email across all devices
 - The actual email stays on the server
- ▼ SMB -Server Message Block (TCP Port 445)
1. Protocol used by Microsoft Windows
 - File sharing, printer sharing
 - Also called CIFS (Common Internet File System)
 2. Using NetBIOS over TCP/IP (NetBT)
 - UDP Port 137 - NetBIOS name service (nbname)
 - UDP Port 139 - NetBIOS session service
 3. Direct over tcp port 445 (NetBIOS-less)
 - Direct SMB communication over TCP without the NetBIOS transport
- ▼ LDAP/LDAPS - Lightweight Directory Access Protocol (TCP Port 389)
- Store and retrieve information in a network directory
 - Commonly used in Microsoft Active Directory
- ▼ RDP - Remote Desktop Protocol (TCP Port 3389)
1. Share a desktop from a remote location over TCP Port 3389
 2. Remote Desktop Services on many Windows versions.
 3. Can connect to an entire desktop or just an application
 4. Clients for Windows, macOS, Linus, Unix, iPhone, Android and others

▼ 2.2 Wireless network technology

▼ Wireless technology

1. IEEE - Institute of Electrical and Electronics Engineers

- 802.11 committee
- Everyone follows these standards

2. Also referenced as a generation

- 802.11ac is WiFi 5
- 802.11ax is WiFi 6 and WiFi 6E
- 802.11be is WiFi 7
- Future versions will increment accordingly

▼ 802.11 technology

1. Frequencies

- 2.4 GHz, 5 GHz, and 6 GHz
- Sometimes a combination

2. Channels

- Group of frequencies, numbered by the IEEE
- Wireless networking channels are governed by country-specific regulations
- Using non-overlapping channels would be optimal

3. Bandwidth

- Amount of frequency in use
- 20 MHz, 40 MHz, 80 MHz, 160MHz, etc

Wi-Fi Gen	IEEE Standard	Release Date	2.4 GHz	5 GHz	Max Data Rate
Wi-Fi	802.11	1997	Yes	No	2 Mbps
Wi-Fi 1	802.11b	1999	Yes	No	11 Mbps
Wi-Fi 2	802.11a	1999	No	Yes	54 Mbps
Wi-Fi 3	802.11g	2003	Yes	No	54 Mbps
Wi-Fi 4	802.11n	2009	Yes	Yes	600 Mbps
Wi-Fi 5	802.11ac	2013	No	Yes	6.93 Gbps
Wi-Fi 6	802.11ax	2019	Yes	Yes	14 Gbps

▼ Bluetooth

1. Remove the wires

- Headsets
- Speakers
- Keyboards

2. Use the 2.4 Ghz range

- Unlicensed ISM (Industrial, Scientific and Medical) band
- Same as 802.11

3. Short-range

- Most consumer devices operate to about 10 meters
-

▼ RFID (Radio Frequency Identification)

1. It's everywhere

- Access badges
- Inventory / Assembly line tracking
- Pet / Animal identification
- Anything that needs to be tracked

2. Radar technology

- Radio energy transmitted to the tag
- RF powers the tag, ID is transmitted back
- Bidirectional communication
- Some tag formats can be active / powered

▼ NFC (Near Field Communication)

1. Two way wireless communication

- Builds on RFID, which is mostly one-way
-

2. Payment systems

- Major credit cards

- Online wallets
- 3. Bootstrap for other wireless
 - NFC helps with Bluetooth pairing
- 3. Access token, identity "card"
 - Short range with encryption support

▼ 2.3 Network Service

▼ DNS server

1. Domain name system
 - Convert names to IP addresses
 - Convert IP addresses to names
2. Distributed naming system
 - The load is balanced across many different servers
 -
3. Usually managed by the ISP or enterprise IT department
 - A critical resource

▼ DHCP server

1. Dynamic Host Configuration Protocol
 - Automatic IP address configuration
2. Very common service
 - Available on most home router
3. Enterprise DHCP server will be redundant
 - Usually running on central server

▼ File share

1. Centralized storage of documents, spreadsheets, videos, picture, and any other file
 - A fileshare
2. Standard system of file management

- SMB (Server Message Block), Apple Filing Protocol (AFP), etc
3. The front-end hides the protocol
 - Copy, delete, rename, etc
- ▼ Print server
1. Connect a printer to the network
 - Provide printing service for all network device
 2. May be software in a computer
 - Computer is connected to the printer
 3. May be built in to the printer
 - Network adapter and software
 3. User standard printing protocols
 - SMB (Server Message Block), IPP (Internet printing protocol), LPD (Line printing daemon)
- ▼ Syslog
1. Standard for message logging
 - Diverse systems, consolidated log
 2. Usually a central logging receiver
 - Integrated into a SIEM (Security information and event manager)
 3. You're going to need a lot of disk space
- ▼ Web server
1. Respond to browser requests
 - Using standard web browsing protocols - HTTP/HTTPS
 - Pages are built with HTML, HTML5
 2. Web pages are stored on the server
 - Downloaded to the browser
 - Static pages or built dynamically in real-time
- ▼ Authentication, Authorization, Accounting (AAA) server
1. Authentication, Authorization, Accounting (AAA) server

- The login server
- 2. Login authentication to resources
 - Centralized management
- 3. Almost always an enterprise service
 - Not required on a home network
- 4. Usually a set of redundant servers
 - Always available
 - Extremely important service
- 5. **RADIUS** (Remote Authentication Dial-In User Service) - AAA solution uses UDP as its transport protocol, combines authentication and authorization, and is commonly deployed for network access control such as VPNs and Wi-Fi authentication
- 6. **TACACS+** (Terminal Access Controller Access-Control System Plus)
 - relies on TCP, separates authentication, authorization, and accounting, and is commonly used for device administration in enterprise networks

▼ Database server

1. Database table storage
 - Save information in columns and rows
 - Similar to a spreadsheet
2. Relational database
 - Link table together
 - These link are relation
 - Flexible and fast
3. Structured Query Language (SQL)
 - Access data using standart language
 - Microsoft SQL Server, MySQL, etc

▼ NTP Server

1. The time of day is more important than you might think

- NTP - Network Time Protocol
- Encryption, logins, backups, log timestamps
- Everything need work from the same clock

2. NTP Server

- Responds to time requests from NTP clients

3. NTP Client

- Requests time updates from NTP server
- Daily synchronization is common

▼ Spam gateways

1. Unsolicited email

- Stop it at the gateway before it reaches the user
- On-site or cloud-based

▼ All in one security appliance

1. Next generation firewall, Unified Threat Management (UTM), Web security gateway

2. URL filter/ Content inspection

2. Malware inspection

3. Spam filter

4. CSU / DSU

5. Router/ Switch

6. Firewall

7. IDS/ IPS

8. Bandwidth shaper

9. VPN endpoint

▼ Load balancers

1. Distribute the load

- Multiple servers

- Invisible to the end-user
2. Large-scale implementations
 - Web server farms, database farms
 3. Fault tolerance
 - Server outages have no effect very fast convergence
 - Very fast convergence
- ▼ Proxy server
1. An intermediate server
 - Client makes the request to the proxy
 - The proxy performs the actual request
 - The proxy provides result back to the client
 2. Useful features
 - Access control, caching, URL filtering, content scanning.
- ▼ SCADA (Supervisory Control and Data Acquisition System) / ICS (Industrial Control System)
1. Supervisory Control and Data Acquisition System (SCADA)
 - Large- scale, multi-site Industrial Control System (ICS)
 2. PC manages equipment
 - Power generation, refining, manufacturing equipment
 - Facilities, industrial, energy, logistics
 3. Distributed control systems
 - Real-time information
 - System control
 2. Requires extensive segmentation
 - No access from the outside
- ▼ Legacy and embedded system
1. Legacy systems
 - Another expression for “really old”

- May also be “really important”
- Learning old things can be just as important as learning the new things
- Outdated, but performs a critical role that is still in use Unable to provide security updates. Are often isolated

2. Embedded system

- Design for a specific purpose
- Not usual to have direct access to the operating system
- Alarm system, door security, time card system
- PLCs - a specific type of embedded system used in machinery

▼ IoT - Internet of Things

1. Appliances

- Refrigerators

2. Smart Devices

- Smart Speakers Respond to voice commands

2. Air control

- Thermostats, temperature control

2. Access

- Smart doorbells, doors

3. May require a segmented network

- Limit any security breaches

▼ 2.4 DNS Configuration

▼ Address records (A) (AAAA)

1. Defines the IP address of host

- This is the most popular query

2. A record are for IPv4 addresses

- Modify the A record to change the host name to IP address resolution

2. AAA record are for IPv6 addresses

- The same DNS server, different record

▼ Canonical name records (CNAME)

1. A name is an alias of another, canonical name

- One physical server, Multiple services

2. The "Nickname" Analogy

- Imagine you host a website on a server with the IP `104.18.32.10`.

1. You create an **A Record** for your main site: `example.com` in `104.18.32.10`.

2. You create **CNAMEs** for other services:

- `www.example.com` = `example.com`
- `shop.example.com` = `example.com`
- `blog.example.com` = `example.com`

▼ Mail exchanger record (MX)

1. Determines the host name for the mail server

- This isn't an IP address; it's a name

▼ Text record (TXT)

1. Human-readable text information

- Originally, these were meant for any human-readable notes, but today they are used for **Security** and **Verification**.

2. DKIM (DomainKeys Identified Mail)

- This adds a "digital signature" to your emails so the receiver knows the message wasn't changed while it was being sent.
- The public key is in the DKIM TXT record

2. SPF (Sender Policy Framework)

- This lists which mail servers are *allowed* to send email for your domain. It prevents hackers from pretending to be you (spoofing).
- Mail servers perform a check to see if incoming mail really did come from authorized host

4. DMARC (Domain based Message Authentication, Reporting and Conformance)

- If **SPF** and **DKIM** (which live inside TXT records) are the ID cards, **DMARC** is the **Security Guard** who decides what to do if those ID cards are missing or fake.
- Accept all, send to spam, or reject the email

▼ 2.4 DHCP

▼ Leases

| DORA four steps of how a device gets a lease

1. Discovery

- Find a DHCP server
- The client (your PC) shouts, *"Is there a DHCP server out there? I need an IP!"*

2. Offer

- Get an offer
- The DHCP server hears the shout and says, *"I have one! Here is 192.168.1.50 for you."*

2. Request

- Lock in the offer
- The client says, *"I'll take it! Please officially assign 192.168.1.50 to me."*

2. Acknowledgment (ACK)

- DHCP server confirmation
- The server says, *"Done! You have a lease for the next 24 hours. Enjoy!"*

▼ Reservations

1. DHCP Reservations (MAC Binding)

- This is a "Magic Trick" in networking. You can tell the DHCP server: *"Whenever you see the laptop with this specific **MAC Address** (Hardware ID), **ALWAYS** give it the IP 192.168.1.50."*

- **Why use this?** It's like having a "Reserved" sign at a restaurant. The device gets the same IP address every time (like a Static IP), but you manage it automatically through the DHCP server instead of typing it in manually on the device.

▼ Scope

1. A **Scope** is simply the **range** of IP addresses that the DHCP server is allowed to hand out.
 - *Example:* Your router might have a scope of 192.168.1.100 to 192.168.1.200 .
 - This means it has **101 addresses** available for guests, phones, and laptops.

▼ Exclusions

1. Sometimes, you want to "carve out" a piece of the scope so the DHCP server **never** gives those numbers away.
 - *Example:* You might exclude 192.168.1.100 to 192.168.1.110 because you want to manually give those addresses to your office printers.

▼ VLANs and VPNs

▼ VLANs (Virtual Local Area Network)

1. A VLAN is used to **segment** (split up) a physical network into smaller, private groups.
2. a logical grouping of computers that allows hosts to communicate as if they are on the same broadcast domain, regardless of their physical location

▼ VPNs (Virtual Private Network)

1. A VPN is used to create a **secure tunnel** over a public network (the internet)
2. encrypted connections between remote locations
2. Remote Access VPN (Client-to-site VPN)
 - This is what most people mean when they say "VPN."

- **How it works:** You install a piece of software on your laptop (like Cisco AnyConnect or OpenVPN). You click "Connect," and just your laptop is now inside the office network.

2. Site-to-Site VPN

- This is used to connect two entire buildings together
- **How it works:** Instead of software on a laptop, the **routers** at both buildings talk to each other and create a permanent "bridge."

▼ 2.5 Networking Hardware Device

▼ Router

1. A router is a device that forwards data packets between different computer networks.

- Connects your home network to the **Internet**
- They care about **IP Addresses**.
- Example : You are setting up a small office. You have 10 computers that need to talk to each other to share files, and they also need to be able to access the internet.

▼ Switches

1. Many ports and features

- The core of an enterprise network
- May provide Power over Ethernet (PoE)

2. Bridging done in hardware

- Forwards traffic based on data link address
- Application-specific integrated circuit (ASIC)

2. Unmanaged Switch

- This is "Plug and Play." You plug it in, and it works. There are no settings to change. It's great for a simple home setup.
- Fixed configuration - No VLANs
- Very little integration with other devices - No management protocols

- Low price point - Simple is less expensive

3. Managed Switch

- VLAN support - Interconnect with other switches via 802.1Q
- Traffic prioritization - Voice traffic gets a higher priority
- Redundancy support - Maintain uptime if a switch fails
- Port mirroring - Capture packets
- External management - Simple Network Management Protocol (SNMP)

▼ Access Point

1. An **Access Point** is not a router.

- It is a device that connects to a wired switch and provides **Wi-Fi** to a specific area.

2. An access point is a bridge

- Extends the wired network onto the wireless network
- Makes forwarding decisions based on MAC address

▼ Patch panels

1. Combination of punch-down blocks and RJ-45 connectors

2. Runs from desks are made once

- Permanently punched down to patch panel

2. Patch panel to switch can be easily changed

- No special tools
- Use existing cables

2. Why use it?

- It's for cable management.
- You connect a short "patch cable" from the Patch Panel to the **Switch**.
- If you need to move a person to a different network, you just move the small cable on the front of the panel instead of re-wiring the whole building.

▼ Firewalls

1. Filters traffic by port number
 - OSI layer 4 (TCP/UDP)
 - Some firewalls can filter based on the application
2. Can encrypt traffic into/out of the network
 - Protect your traffic between sites
2. Can proxy traffic
 - A common security technique
2. Most firewalls can be layer 3 devices (routers)
 - Usually sits on the ingress/egress of the network
3. Firewalls
 - **Network-based Firewall - A physical box (or a feature inside your router) that protects the entire house/office.**
 - **Host-based Firewall** - Software running on your specific PC (like Windows Defender Firewall) that protects only that one computer.

▼ Power over Ethernet (PoE)

1. Power provided on an Ethernet cable
 - One wire for both network and electricity
 - Phones, cameras, wireless access points
 - Useful in difficult-to-power areas
2. Power provided at the switch
 - Built-in power - Endspans
 - In-line power injector - Midspans
2. PoE Injectors
 - What if you have a PoE camera, but your switch is old and doesn't support PoE? You use a PoE Injector.
 - It's a small "middle-man" box that you plug into a wall outlet. It "injects" power into the Ethernet cable before it reaches the

camera.

2. PoE

- The original PoE specification
- 15.4 watts DC power, 350 mA max current

3. PoE+

- 25.5 watts DC power, 600 mA max current

3. PoE++

- 51 W(Type 3), 600mA max current
- 71.3 W (Type 4), 960 mA max current
- PoEwith 10GBASE-T

3. Cable broadband

- the use of cable modems for Internet access over a standard cable television infrastructure

▼ Cable modem

1. Broadband

- Transmission across multiple frequencies
- Different traffic types

2. Data on the "cable" network

- DOCSIS - (Data Over Cable Service Interface Specification)

2. High-speed networking

- Speeds up to 1 Gigabit/s are common

2. Multiple services

- Data, voice, video

▼ DSL modem - (Digital Subscriber Line)

1. DSL (Digital Subscriber Line)

- Uses telephone lines

2. Download speed is faster than the upload speed (asymmetric)

- 200 Mbit/s downstream / 20 Mbit/s upstream are common

- '10,000 foot limitation from the central office (CO)
- Faster speeds may be possible if closer to the CO

▼ ONT - Optical network terminal

1. ONT - Optical network terminal
 - Fiber to the premises
2. Connect the ISP fiber network to the copper network
 - Demarcation point (demarc) in the data center
 - Terminal box on the side of the building
2. Line of responsibility
 - One side of the box is the ISP
 - Other side of the box is your network

▼ Network Interface Card (NIC)

1. The fundamental network device
 - Every device on the network has a NIC
 - Computers, servers, printers, routers, switches, phones, tablets, cameras, etc.
2. Specific to the network type
 - Ethernet, WAN, wireless, etc.
2. Often built-in to the motherboard
 - Or added as an expansion card
2. Contains the hardware address
 - Media Access Control (MAC) address
 - A unique hardware designation

▼ 2.6 IPv4 and IPv6

▼ IPv4

1. An IPv4 address consists of 32 bits
2. IPv4 addresses are expressed with the use of Decimal numbers

2. Private IP addresses

- Assigned within a home or business network for internal communication
- Cannot be routed over the public Internet and are only valid within the confines of a local network
- Can be reused across different networks (i.e., they do not need to be unique globally)

2. Private IP addresses Common Ranges

- **10.0.0.0 - 10.255.255.255 (10.0.0.0/8)** (Large corporations) - Class A range
- **172.16.0.0 - 172.31.255.255 (172.16.0.0/12)** - Class B range
- **192.168.0.0 - 192.168.255.255 (192.168.0.0/16)** (Most home routers) - Class C range

3. Public IP addresses

- Assigned by ISPs to devices for communication over the Internet
- Unique across the entire Internet, ensuring that devices can be identified and located globally
- Used for devices that need to communicate directly with the Internet and are routable through the global internet infrastructure

▼ IPv6

1. An IPv6 address consists of 128 bits
2. IPv6 addresses are expressed with the use of Hexadecimal numbers
2. IPv6 was primarily developed to mitigate the issue of Address exhaustion
2. **No more Private/Public split:** Every device can technically have its own unique global address.
3. **Link-Local Address:** Every IPv6 device automatically gives itself an address starting with **fe80::**. This allows it to talk to

other devices on the same local switch without a router or DHCP server.

▼ Automatic Private IP Addresses (APIPA)

1. APIPA allows a Windows host to self-configure an IPv4 address and subnet mask when a DHCP server is unavailable.
2. APIPA uses the address block range from 169.254.0.0 to 169.254.255.255.
2. APIPA-assigned addresses are valid only for communication within the network segment to which the host is connected.
2. A host with an APIPA-assigned address cannot connect to the Internet.

▼ Static IP addresses

1. Provide a permanent address, essential for services requiring consistent reachability
2. Must be manually assigned or reserved through administrative tools, requiring more management
2. Ideal for servers, network devices needing a stable address, and remote access applications

▼ Dynamic IP addresses

1. Change periodically, making them unsuitable for hosting services that require a fixed address
2. Assigned automatically by a DHCP server, minimizing user configuration
2. Commonly used for consumer devices, prioritizing ease of use, security, and cost efficiency

▼ Subnet mask

1. a 32-bit number that divides
2. It identifies which part of an IP address belongs to the network and which part belongs to the host
2. *Example:* 255.255.255.0 tells the PC that the first three numbers of the IP must match everyone else on the network.

▼ Default Gateway

1. A network device (router) that enables exchange of data between hosts residing in different subnets.

▼ 2.7 Network Connection Type

▼ Satellite networking

1. Communication to a satellite
 - Non terrestrial communication
2. High cost relative to terrestrial networking
 - 100 Mbit/s down, 5 Mbit/s up are common
 - Remote sites, difficult to network sites
2. Relative high latency
 - Some systems have 250 ms up, 250 ms down
 - Starlink advertises 25 to 60 ms
2. Not always a perfect option
 - Line of sight, rain fade

▼ Fiber

1. High speed data communication
 - Frequencies of light
2. Higher installation cost than copper
 - Equipment is more costly
 - More difficult to repair
 - Communicate over long distances
2. Large installation in the WAN core
 - Supports very high data rates
 - SONET, Wavelength division multiplexing
2. Fiber has appeared at the premises
 - Business and home use

▼ Cable

1. Broadband

- Transmission across multiple frequencies
- Different traffic types

2. Data on the "cable" network

- DOCSIS (Data Over Cable Service Interface Specification)

2. High - speed networking

- 50 Mbits/s through 1000+Mbit/s are common

2. Multiple services

- Data, voice, video

▼ DSL

1. ADSL (Asymmetric Digital Subscriber Line)

- User telephone lines

2. Download speed is faster than the upload speed (asymmetric)

- 200 Mbit/s downstream/ 20 Mbit/s upstream are common
- '10,000 foot limitation from the central office (CO)
- Faster speeds may be possible if closer to the CO

▼ Cellular networks

1. Mobile Devices

- "Cell" phones

2. Separate land into "cells"

- Antenna covers a cell with certain frequencies

3. Tethering

- Turn your phone into a wireless router

4. Mobile hotspot

- Standalone devices
- Use your phone for other things

▼ WISP (Wireless Internet Service Provider)

1. Wireless Internet Service Provider

- Terrestrial internet access using wireless
- 2. Connect rural or remote locations
 - Internet access for everyone
- 3. Many different deployment technologies Meshed 802.11
 - Meshed 802.11
 - 5G home internet
 - Proprietary wireless
- 4. limitations of WISP-based connectivity
 - Clear line-of-sight dependency
 - Susceptibility to RF interference and environmental factors
 - Limited bandwidth capacity relative to wired broadband
 - Signal degradation over longer distances

▼ 2.7 Network Type

▼ LAN - Local area network

1. Local area network
 - Local is relative
2. A building or group of buildings
 - High-speed connectivity
3. Ethernet and 802.11 wireless
 - Any slower and it isn't "local"

▼ WAN - Wide area network

1. WAN - Wide area network
 - Spanning the globe
2. Generally connects LANs across a distance
 - And generally much slower than the LAN
3. Many different WAN technologies
 - Point-to-point serial, Multiprotocol Label Switching (MPLS), etc.

▼ PAN - Personal area network

1. Key Characteristics

- **Short Range:** Typically covers a radius of about **10 meters (30 feet)**.
- **Low Power:** Because the devices are so close, they don't need high-powered transmitters. This saves battery life on mobile devices.
- **Ad-hoc:** Devices often join and leave a PAN dynamically (like your headphones connecting to your phone as soon as you put them on).

2. Common Technologies

- **Bluetooth:** The most common PAN technology. It uses the 2.4 GHz frequency to link headsets, keyboards, mice, and game controllers
- **USB (Universal Serial Bus):** Even a wired connection between your PC and your phone or a printer is technically a "Wired PAN."

▼ MAN - Metropolitan Area Network

1. Key Characteristics

- **Geographic Scope:** Typically covers a city, a large town, or a large campus (usually a 5km to 50km radius).
- **Ownership:** Often owned by a single large entity (like a city government or a university) or a service provider that rents the connection to businesses.
- **High Speed:** Because it uses fiber optic cables to connect sites, it is much faster than a standard internet connection.

2. Common Technologies

- **Metro Ethernet:** Using standard Ethernet technology but over a massive scale to connect different parts of a city.
- **Fiber Optics:** The backbone of most MANs, allowing data to travel between buildings at light speed.
- **Microwave/Radio Links:** Sometimes used to connect buildings where digging trenches for cables is too difficult (like across a

river or a busy highway).

▼ Storage area network (SAN)

1. Key Characteristics

- **Block-Level Access:** Unlike a NAS (which handles files), a SAN handles "blocks" of data. This makes it much more efficient for databases and high-end applications like video editing or 3D rendering.
- **Redundancy:** SANs are designed so that if one cable or one switch fails, the data keeps flowing.
- **High Performance:** It uses specialized technology to ensure there is zero "lag" between the server and the hard drive.

2. Common SAN Technologies

- **Fibre Channel (FC):** A very expensive, very fast fiber optic technology dedicated specifically to storage. It requires special "Host Bus Adapters" (HBAs) instead of regular network cards.
- **iSCSI (Internet Small Computer Systems Interface):** A way to run SAN commands over regular Ethernet cables. It's cheaper than Fibre Channel and very popular in medium-sized businesses.

▼ Wireless local area network (WLAN)

1. Key Characteristics

- **Standard:** WLANs use the IEEE 802.11 standards, which we commonly call Wi-Fi.
- **Flexibility:** The biggest advantage is mobility. You can move around the office or house without losing your connection.
- **Range:** Typically covers a few hundred feet, though this can be blocked by walls, metal, or water.

2. The Core Components

- **Wireless Access Point (WAP or AP):** This is the central hub. It connects to the wired network and "broadcasts" the signal to the air.

- Wireless Network Interface Card (NIC): Every device (your laptop, smartphone, or ROG Ally) has a small wireless radio and antenna inside it that allows it to talk to the Access Point.

▼ 2.8 Network tools

▼ Cable crimpers

a crimper tool is used for attaching connectors onto cables by compressing the connector's metal contacts onto the exposed wires.

1. "Pinch" the connector onto a wire
 - Coaxial, twisted pair, fiber
2. Connect the modular connector to the Ethernet cable
 - The final step of the process
3. Metal prongs are pushed through the insulation
 - The plug is also permanently pressed onto the cable sheath



▼ Cable stripper

removing electrical insulation cover from electric wires



▼ Wi-Fi analyzer

1. Measuring the strength of the Wi-Fi signal
2. Detecting interference from other devices or networks
3. Analyzing wireless channel usage
4. View Wi-Fi information
 - Frequencies/channels, signal strength access points, interference, wireless devices



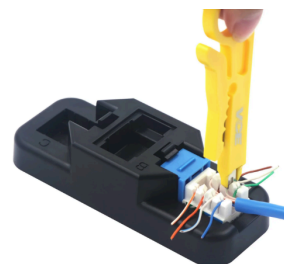
▼ Toner probe

1. Toner & probe kit
 - trace and identify individual wires within a cable bundle
2. Tone generator
 - Puts an analog sound on the wire
3. Connect the tone generator to the wire
 - Modular jack
 - Coax
 - Punch down connectors



▼ Punch down tool

1. used for attaching network cables to a patch panel



▼ Cable tester

1. verify the integrity and connectivity of network cables
2. Can identify missing pins Or crossed wires

▼ Loopback plug

1. Used to test the functionality of a NIC
2. Useful for testing physical ports



▼ Network tap

1. enables network traffic monitoring without interrupting the data flow



3 Hardware

▼ 3.1 Display type

▼ LCD (Liquid Crystal Display)

a display technology most commonly used in modern computing devices

1. Twisted Nematic (TN)

- The original LCD technology
- Low color quality
- Low viewing angles - color shifts
- Fast response times

2. In-Plane Switching (IPS)

- High color quality
- Wide viewing angles
- Fast response times
- Can be more expensive to produce than TN

3. **Vertical Alignment (VA)**

- Good color quality
- Good viewing angles
- High contrast ratios
- Slow response times
- A good compromise between TN and IPS

▼ OLED (Organic Light Emitting Diode)

1. Organic compound emits light when receiving an electric current
2. Provides exceptional contrast ratio and black levels
3. No backlight
 - The organic compound provides the light
4. Thinner and lighter
 - Flexible and mobile- no glass needed
 - Enables lightweight, bendable, and ultra-slim screens
5. Tablets, phones, smart watches
 - Very accurate color representation
 - Tends to be a bit higher cost than LCD
6. **The Trade-off:** OLEDs can suffer from "**Burn-in**" if a static image (like a taskbar or a game's HUD) is left on the screen for thousands of hours.

▼ Mini-LED (Mini light-emitting diode)

1. Same backlight technology as a conventional LED-backlit LCD, But much smaller Light Emitting Diodes (LEDs)
2. Each LED can be enabled or disabled And color and intensity can be different

3. Much better control over dark screen areas Deeper blacks, better color representation
4. Offer excellent contrast due to refined backlight control
5. Achieve high peak brightness suitable for bright environments

▼ Touch screen/digitizer

1. capable of performing both input and output functions
2. A mobile device component that translates analog data into a format suitable for computer processing
3. Merge laptop and tablet input
 - Digitizer responds to touch
 - No keyboard required
 - But often still available
4. A mobile device component responsible for taking input
5. Use a pen-like device as input
 - Stylus input
 - Useful for graphical input
6. Used commonly on laptop / tablets
 - Or hybrid devices

▼ Inverter

1. **It turns low-voltage DC power into high-voltage AC power.**
 - **The Motherboard** provides low-voltage **DC** (Direct Current).
 - **The CCFL Bulb** requires high-voltage **AC** (Alternating Current) to ignite the gas and glow.
 - **The Inverter** sits in the middle, acting as a "power converter" to bridge that gap.
2. LED no Inverter
 - LEDs run on low-voltage **DC** power, which the motherboard already provides.

- Because there's no need to convert to high-voltage AC, the inverter has been replaced by a much simpler **LED Driver** circuit.

▼ 3.1 Display attribute

▼ Pixel density

1. The number of individual pixels packed into a given area of the screen
2. Measure how many pixels in an inch
 - Or pixels per centimeter
 - Apply a physical measurement to a display
3. A higher pixel density looks better
 - Higher clarity and sharpness
4. Consider how the image will be viewed
 - Displays
 - Printers (dots per inch)

▼ Refresh rate

1. The number of times the screen updates the image per second
2. Usually measured in hertz (Hz)/ Frames per second (FPS)
3. Frame rates may Vary
 - Movies: 24 fps
 - Television: 30 fps
 - Video games: 60 fps and higher

▼ Screen resolution

1. The total number of pixels displayed on the screen
2. The number of pixels
3. Many standards
 - **720:** 1280×720
 - **HD:** 1920×1080
 - **1440:** 2560×1440

- **4K:** 3840×2160

▼ Color gamut

1. The range of colors a display can reproduce
2. A display may be compared with an existing color gamut standard
3. Gamut compatibility can vary widely
 - Higher percentages tend to be more expensive
4. OLED displays provide a wide color gamut

▼ 3.2 Network Cable

▼ Copper

1. T568A /T568B standards (twisted-pair cables)

twisted-pair cabling types is that wires inside the cable are grouped into pairs, and the wires in each pair are twisted around each other to reduce signal interference from adjacent wire pairs

- Define the pinout configuration for RJ45 connectors used in Ethernet cabling
- Specify the pin assignments for the eight wires contained in twisted-pair cables

2. Coaxial

- Commonly used for cable television signals
- Features a single copper conductor surrounded by insulation and shielding
- Classified as a copper-based transmission medium
- Typically used in broadband cable Internet connections

3. Shielded twisted pair (STP)

- Takes advantage of an additional protective cover reducing signal interference from outside sources



STP Cable

4. Direct burial STP

- Enhanced shielding
- Underground network infrastructure
- Water and chemical resistance
- Outdoor network infrastructure
- Durable outer jacket



5. Unshielded twisted pair (UTP)

- Classified as twisted-pair copper cabling



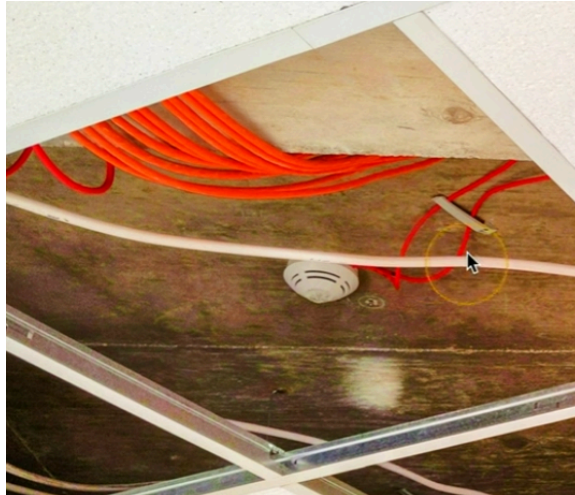
UTP Cable

▼ Plenum-rated 阻燃级电缆

1. plenum space

- A type of enclosed space in a building

- between a dropped ceiling and the structural ceiling
 - used for air handling
2. A special type of cabling with a fire-retardant jacket, designed for use in that space, is known as plenum-rated cabling.



▼ Optical

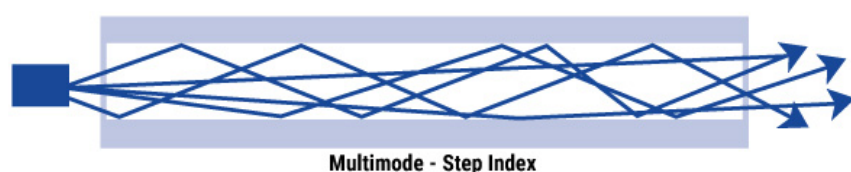
1. Single-mode fiber optics

- Typically incurs a higher cost than multimode fiber optic systems
- Offers long-distance transmission capabilities, often exceeding 100 km
- Utilizes laser light as the source of transmission



2. Multimode fiber optics

- Supports transmission distances of up to 2 km
- Usually less expensive than single-mode fiber optics
- Employs LED light sources for transmission



▼ Peripheral cables

▼ USB - Universal Serial Bus

USB Version				Bandwidth	Data Transfer Speed	Logo	Connector
Official release	Slogan	Marketing name					
USB4	V1.0	USB4 20Gbps USB4 40Gbps	USB4	20 or 40Gbps	20GBps 40GBps		
USB 3.2	Gen 2x2	SuperSpeed+	USB 3.2	20Gbps	25GBps		
	Gen 2	SuperSpeed+	USB 3.1	10Gbps	12.8GBps		
	Gen 1	SuperSpeed	USB 3.0	5Gbps	6.4GBps		
USB	2.0	High-Speed	USB 2.0	480Mbps	60MBps		
	1.1	Full Speed	USB 1.1	12Mbps	1.5MBps		
	1.0		USB 1.0	1.5Mbps	0.192MBps		

Standard	Year	Logo	Data transfer speed	Cable length
USB 1.1	1998		12 Mbps	3m
USB 2.0	2000		480 Mbps	5m
USB 3.0	2008		4.8 Gbps	3m
USB 3.1 Gen1	2013		5 Gbps	3m
USB 3.1 Gen2	2013		10 Gbps	3m

1. USB 2.0

- the maximum data transfer rate: 480 Mbps
- the maximum allowable cable length: 5 meters

2. USB 3.0

- Backward compatibility with earlier USB versions
- 5 Gbps data transfer rate
- SuperSpeed USB

3. USB 3.1

- 10 Gbps data transfer rate

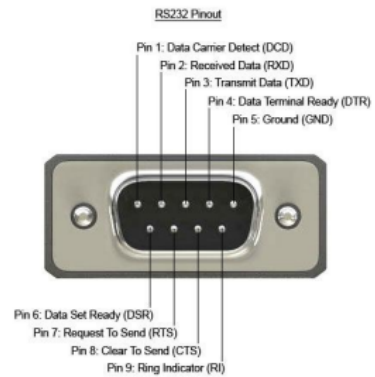
4. USB 3.2

- 20 Gbps data transfer rate

▼ Serial

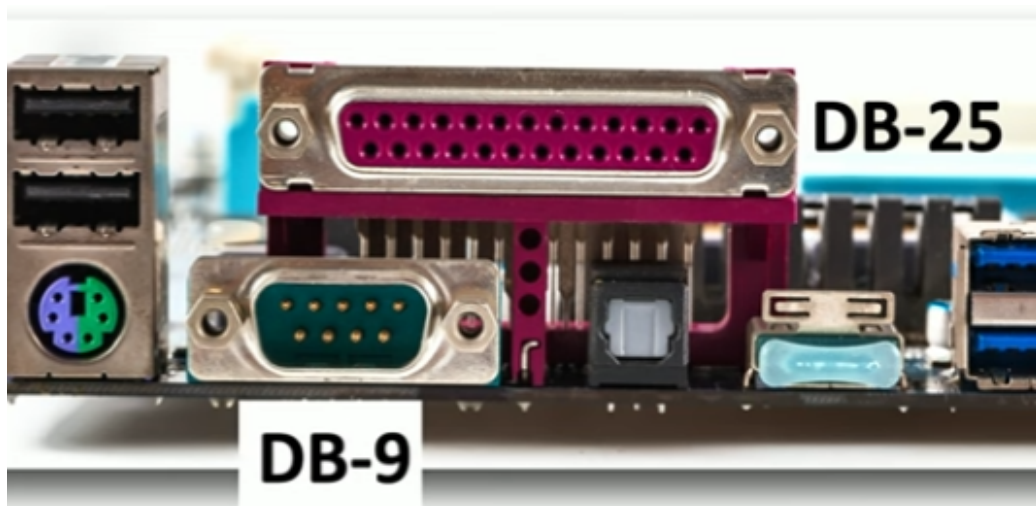
1. RS-232

- Type of serial cable used to connect modems, printers, mice, and other peripheral devices
- An industry standard since 1969



2. D-subminiature or D-sub

- The letter refers to the connector size



▼ Thunderbolt

1. Thunderbolt 1

- uses two separate 10 Gbps data channels
- providing a combined maximum throughput of 20 Gbps

2. Thunderbolt 2

- unified 20 Gbps connection

3. Thunderbolt 3

- overall throughput to 40 Gbps

4. Thunderbolt copper cable

- Up to 3 meters
- 60 meters (fiber-optic Thunderbolt)
- Daisy-chain up to 6 devices

Specification & Development of USB/Thunderbolt www.chargerlab.com					
Protocol	Year	Interface	Speed	Power	Video
USB2.0	2000	USB-A, USB-B, Mini-A, Mini-B, Micro-A, Micro-B	480Mbps	5V 500mA	N/A
USB3.0 USB3.1Gen1 USB3.2 Gen1	2008	USB-A, USB-B, Mini-B, Micro-B	5Gbps	5V 900mA	N/A
USB3.1 USB3.1Gen2 USB3.2 Gen2	2013	USB-A, USB-C	10Gbps	5V 900mA	N/A
USB3.2 USB3.2Gen2X2	2017	USB-A, USB-C	20Gbps	5V 900mA	N/A
USB4	2019	USB-C	40Gbps	20V5A	Single & Dual 4K Single 5K
Thunderbolt 1	2011	Mini-DP	10Gbps	5V2.4A	Single 4K
Thunderbolt 2	2013	Mini-DP	20Gbps	5V2.4A	Single 4K
Thunderbolt 3	2015	USB-C	40Gbps	20V5A	Single & Dual 4K Single 5K
Thunderbolt 4	2020	USB-C	40Gbps	20V5A	Single & Dual 4K Single 8K

▼ Video cables

1. High-definition Multimedia Interface (HDMI)

- transmission of both video and audio data
- Digital signal

2. DisplayPort (DP)

- transmission of both video and audio data
- Digital signal

3. Digital Visual Interface (DVI)

- No audio support
- DVI-A: Analog signal
- DVI-D: Digital signal
- DVI-I: Integrated - Digital and analog in the same connector

4. Video Graphics Array (VGA)

- No audio support
- Analog signal

5. USB-C

- allows it to transmit video signals using protocols such as DisplayPort or HDMI
- deliver video to external displays like monitors or TVs, offering the same functionality as dedicated video cables
- To connect to most external displays, USB-C requires a compatible adapter or cable, such as USB-C to DisplayPort or USB-C to HDMI

▼ Serial Advanced Technology Attachment (SATA)

- SATA interface specification defines a power connector consisting of **15 pins**
- SATA interface specification defines a data cable connector consisting of **7 pins**
- A single SATA data cable can be used to connect a motherboard slot with **Only 1 device**

1. SATA revision 1.0 (SATA I)

- 1.5 Gbps data transfer rate
- Maximum cable length of 1 meter

2. SATA revision 2.0 (SATA II)

- 3 Gbps data transfer rate
- Maximum cable length of 1 meter

3. SATA revision 3.0 (SATA III)

- 6 Gbps data transfer rate
- Maximum cable length of 1 meter

4. External SATA (eSATA)

- 6 Gbps data transfer rate
- Maximum cable length of 1 meter

▼ cable adapters

1. Signal conversion
2. Connector form factor adjustment



▼ 3.3 RAM - Random Access Memory

▼ Notch on the RAM module

- Ensures proper alignment of the module with the memory bank on the motherboard
- Prevents the installation of incompatible memory types.

▼ SODIMM - Small Outline Dual In-line Memory Module

1. Common memory module form factor used in laptops
2. Pin Count: DDR3 SO-DIMM usually has 204 pins, while DDR4 has 260 pins.



▼ DIMM - Dual In-line Memory Module

1. RAM features separate electrical contacts on each side of the module
2. They support modern RAM technologies like DDR3, DDR4, and DDR5, with pins on both sides allowing for independent connections to memory chips



▼ SRAM - Static Random-Access Memory

1. Faster than Dynamic Random-Access Memory (DRAM)
2. More expensive in comparison to Dynamic Random-Access Memory (DRAM)
3. Volatile storage media type
4. Utilized for CPU cache memory chips

▼ DRAM - Dynamic Random-Access Memory

1. Slower than Static Random-Access Memory (SRAM)
2. Less expensive in comparison to Static Random-Access Memory (SRAM)
3. Volatile storage media type
4. Widely used as the primary storage media (regular RAM modules installed in memory slots on the motherboard)

▼ SDRAM - Synchronous Dynamic Random-Access Memory

1. a broad category of DRAM modules that rely on the signal sent by the system clock in order to coordinate their functioning with other internal PC components
 - Queue up one process while waiting for another
 - Classic DRAM didn't Wait for a clock signal
 - No backward compatibility

▼ DDR - Double Data Rate

1. DDR3 SDRAM - Double Data Rate 3 Synchronous Dynamic Random-Access Memory
 - 16 GB maximum capacity per memory module
 - 240 contact pins

- A notch on the RAM module contact surface prevents the installation of incompatible memory types
 - No backward compatibility
2. DDR4 SDRAM - Double Data Rate 4 Synchronous Dynamic Random-Access Memory
 - 32 GB maximum capacity per memory module
 - 288 contact pins
 - A notch on the RAM module contact surface prevents the installation of incompatible memory types
 - No backward compatibility
 3. DDR5 SDRAM - Double Data Rate 5 Synchronous Dynamic Random-Access Memory
 - 64 GB maximum capacity per memory module
 - 288 contact pins
 - A notch on the RAM module contact surface prevents the installation of incompatible memory types
 - No backward compatibility

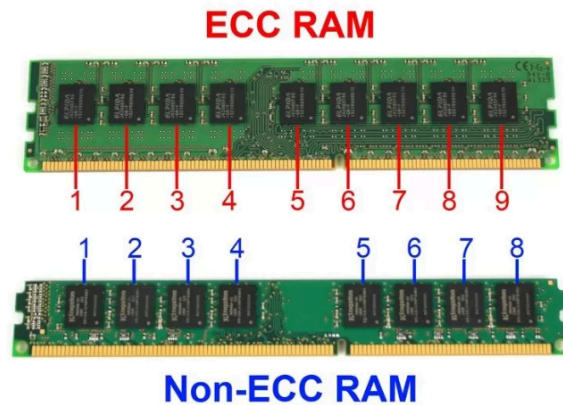
▼ Parity bit

1. Certain types of RAM use an additional bit to detect whether a data error has occurred
2. This extra bit stores information about the count of bits set to 1 in a given data string
3. Examining the value of this bit at different stages of processing allows for the detection of data corruption.

▼ Error-correcting code (ECC)

1. ECC type RAM
 - Can detect and correct errors
 - prioritizes data integrity over cost and performance
 - more expensive and slightly slower than non-ECC RAM
2. Main system memory in desktop PCs

- Non-ECC memory
- Dynamic RAM



▼ Multi-channel memory

1. The color-coded memory slots on the motherboard indicate that a given motherboard provides support for the multi-channel memory architecture

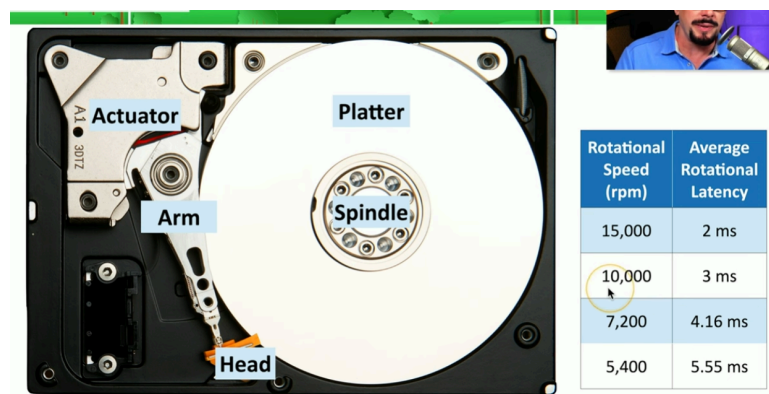


2. Multi-channel memory architecture requires:
 - Memory modules of matching types
 - Memory modules of the same capacity
 - Installing modules in appropriate memory slots (slots of matching color) on the motherboard
 - Memory modules of matching speeds

▼ 3.4 Storage Device

▼ HDD - Hard Disk Drive

1. The platters in a magnetic hard drive spin at a rate measured in Revolutions per minute (RPM)
2. maximum RPM value available in modern HDDs 15000 rpm
3. Non-volatile magnetic storage
4. Form factors
 - 2.5 inch
 - 3.5 inch



▼ SSD - Solid State Drive

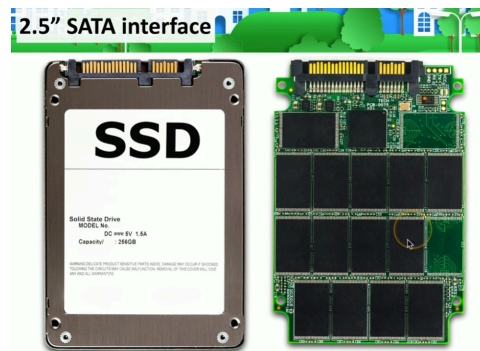
1. NVMe - Non-Volatile Memory Express
 - Developed specifically for SSDs
 - Uses the PCIe interface for data transfer
 - Lower latency, supports higher throughputs, transfer speed of 20 Gbps



2. SATA

- Common in 2.5-inch SSDs replacing traditional HDDs
- Widely used in consumer-grade laptops and desktops

- An older interface originally designed for mechanical hard drives
- Capped at a maximum data transfer rate of 6 Gbps



3. Peripheral Component Interconnect Express (PCIe)

- PCIe - speed interface that serves as the backbone for NVMe SSDs and other high-performance hardware devices.

4. SAS - Serial Attached SCSI (Small Computer System Interface) Interface

- Designed for systems requiring high redundancy, error correction, and uptime
- Optimized for multi-drive configurations in a daisy-chain topology
- Commonly implemented in mission-critical environments like servers and data centers



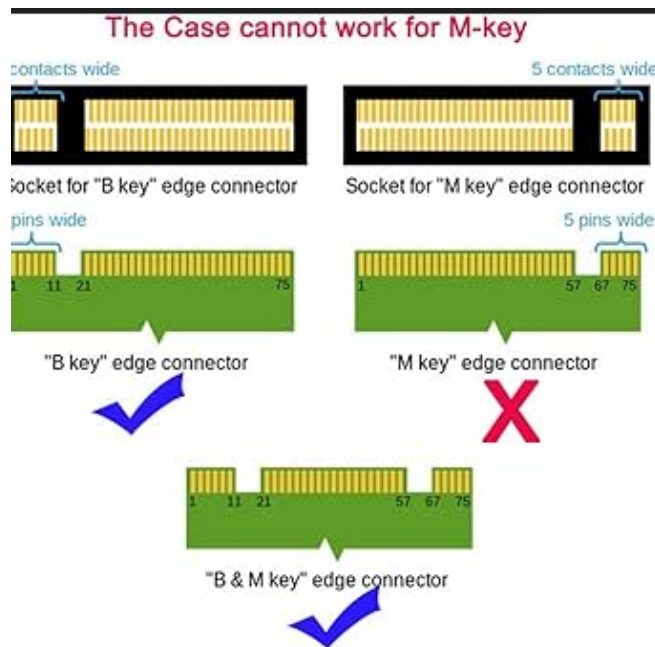
▼ SSD Form Factors

1. M.2

- An M.2 key is a notch on the pin contact surface of an M.2 expansion card which prevents its insertion into an incompatible

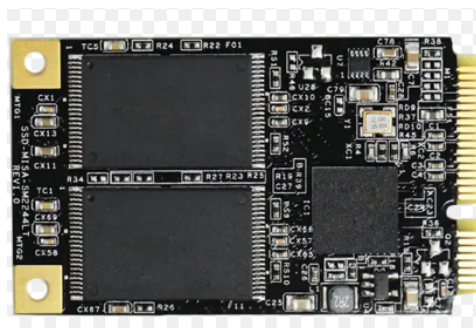
socket

- M.2 expansion cards used for solid-state storage applications have key IDs of B and M
- The B-keyed M.2 SSD cards take advantage of 2 lanes of a PCIe link (lower read/write speed)
- The M-keyed M.2 SSDs use 4 PCIe lanes (higher read/write speed)
- The (B + M) keyed increase their compatibility, as they can be installed in either of the two types of expansion slots on the motherboard.



2. mSATA - Mini-serial Advanced Technology Attachment

- portable device SSD form factor superseded by M.2
- Smaller than 2.5" SATA drives



▼ Removable storage

1. Flash drives

- Plug-and-play across most devices supporting USB
- Designed for general purpose file transfer and storage
- Solid-state memory technology (containing no mechanical moving parts)
- Non-volatile, rewritable storage media type



2. Memory cards

- Compatible with dedicated slots or card readers
- Primarily designed to expand storage capacity of compact electronics
- Solid-state memory technology (containing no mechanical moving parts)
- Non-volatile, rewritable storage media type



▼ Optical Drive

1. Small bumps read with a laser beam
 - Microscopic binary storage
2. Many different formats
 - CD-ROM, DVD-ROM, Blu-ray

▼ 3.4 RAID - Redundant Array of Independent Disks

▼ RAID 0

1. Requires a minimum of 2 drives to implement
2. Is also known as disk striping
3. Decreases reliability (failure of any disk in the array results in the loss of all data in the array)
4. Is suitable for systems where performance takes precedence over fault tolerance

▼ RAID 1

1. Requires at least 2 drives to implement
2. Is also referred to as disk mirroring
3. Improves reliability by duplicating data on each drive (if one drive fails, data remains accessible)

▼ RAID 5

1. Requires at least 3 drives to implement
2. Offers both increased performance and fault tolerance (a single drive failure does not result in data loss; the array can rebuild the missing data)
3. Is also known as disk striping with parity

▼ RAID 6

1. Requires at least 4 drives to implement
2. Offers increased performance and fault tolerance (up to 2 drive failures can be tolerated without data loss)
3. Is also known as disk striping with double parity

▼ RAID 10

1. Requires a minimum of 4 drives to implement
2. Offers increased performance and fault tolerance (the array can survive the failure of one drive in each mirrored pair)

3. Is known as a stripe of mirrors, combining RAID 1 (mirroring) with RAID 0 (striping)

▼ 3.5 Motherboard

▼ Motherboard Form Factor

1. ATX - Advanced Technology Extended

- most common motherboard form factor used in desktop PCs
- up to seven PCIe expansion slots and four DIMM (RAM) slots.



1. mATX - microATX

- the optimal choice for a compact or budget-friendly desktop build
- It fits up to four expansion slots and is compatible with micro-ATX and standard ATX cases



2. ITX - Information Technology eXtended

- form factors is the smallest
- only one PCIe slot and is designed for highly compact cases



▼ Motherboard connector types

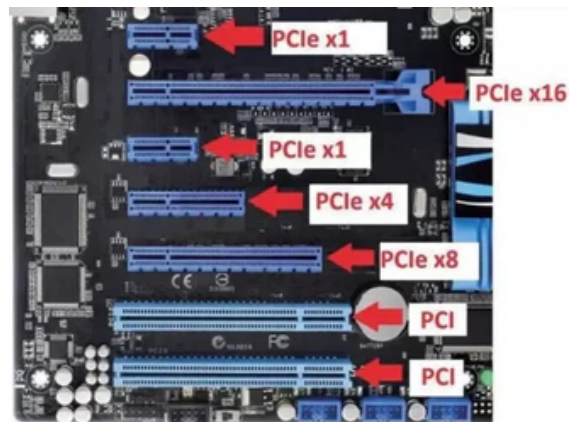
1. PCI - Peripheral Component Interconnect

- Provided a 32-bit or 64-bit bus
- connect hardware components
- such as sound cards, graphics cards, and network adapters—to a computer's motherboard



2. PCIe- Peripheral Component Interconnect Express

- The modern standard (Replaces the older PCI standard)
- providing significantly higher speed than conventional PCI
- scalable performance through lanes (x1, x4, x8, x16, x32)
- with modern generations (Gen 4/5/6)



3. Power Connectors

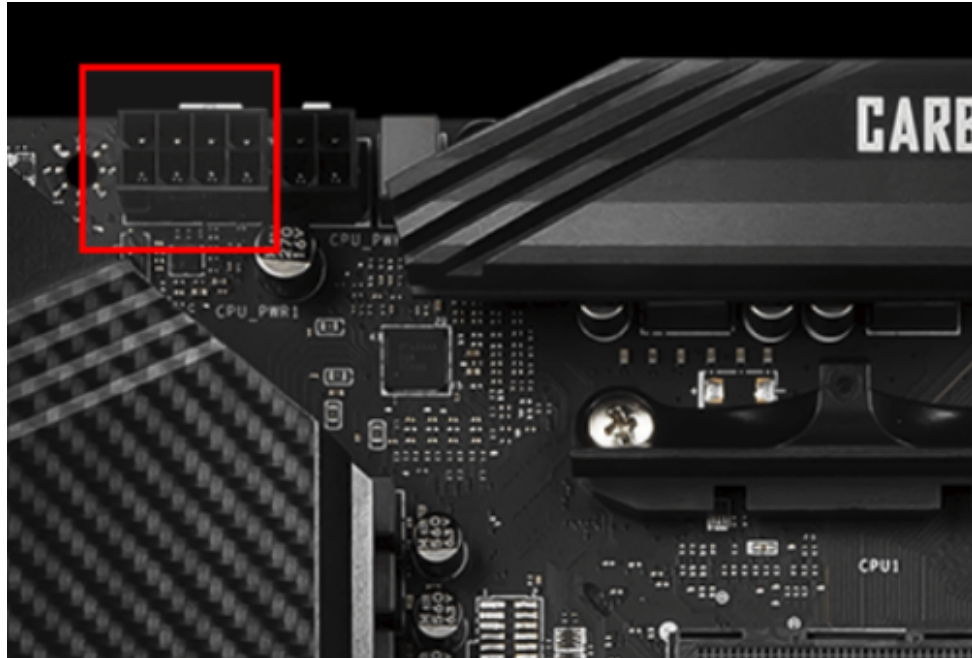
- Main power connector - 24-pin (20-pin original, then add 24 pin for PCIe power)



- PCIe power connector - 6-pin and 8-pin PCIe connector



- CPU power connector - 4-pin (Low)/8 or 4+4-pin (common)/ 12 or 8+4 /16 or 8+8 -pin (Gaming high end) 12V connector



- SATA connector - L-shape design that prevents its improper insertion

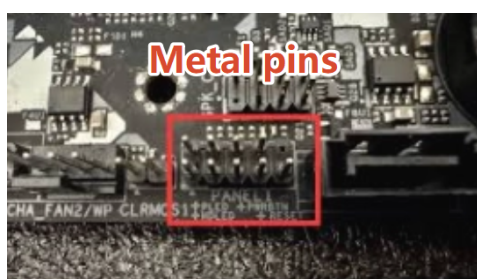


- eSATA connector - **connect external storage devices to a computer at speeds matching internal SATA drives**

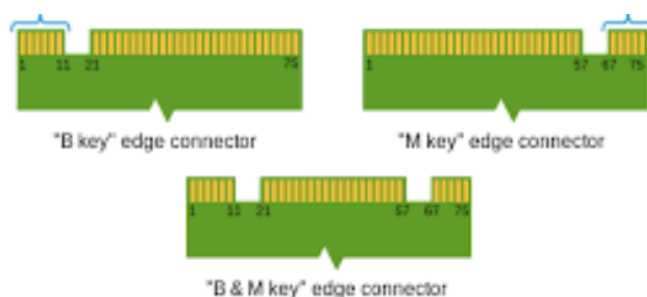


- Headers connector (Metal pins)- sets of pins on a motherboard used to connect various internal components and case features

like power buttons, USB ports, Front Panel (power, reset, status LED), and fans.



- M.2 connector -
 - **M Key:** The standard for high-performance **NVMe SSDs**, **supports up to 4 PCIe lanes for maximum speed**
 - **B Key:** Used for older or slower SATA-based M.2 SSDs and cellular modems. supports PCIe x2 lanes.
 - **B+M Key:** A hybrid design with two notches. These drives can fit into either B or M sockets but are limited to the slower of the two speeds (SATA or PCIe x2)



▼ 3.5 Motherboard compatibility

▼ CPU socket types

1. Advanced Micro Devices, Inc. (AMD)

- **AMD is generally superior for gaming and efficiency**

- AMD-V - AMD's CPU hardware virtualization

2. Intel

- Intel often excels in raw speed for productivity and workstation tasks
- VT-x - Intel CPU hardware virtualization

▼ Server motherboard

1. Multisocket

- Supports multiple physical CPU packages
- Split the load

2. Memory slots

- Supports 4+ modules
- Supports ECC RAM

3. Expansion slots

- Many slots and different sizes

4. Overall size

- Designed for limited space (rack-mount / blade)

▼ BIOS/Unified Extensible Firmware Interface (UEFI) settings

1. BIOS - Basic Input/Output System (Legacy)

- The firmware stored on a motherboard chip that initializes
- Tests (POST) - Power-On Self-Test
- Manages computer hardware during startup
- Boot loader - Start the operating system

2. UEFI BIOS Unified Extensible Firmware Interface

- a modern, secure, and fast firmware standard
- replacing the legacy BIOS

3. Boot Option

- **Common Boot Menu Keys:** Frequently used keys to interrupt startup include **F12**, **F11**, **Esc**, or **Del**, depending on the manufacturer.
 - **Startup Settings (Troubleshooting):** Provides options like Safe Mode (networking/command prompt), Disable Driver Signature Enforcement, and Disable Early Launch Anti-Malware.
 - **BIOS/UEFI Setup:** The foundational menu for changing the boot order, enabling/disabling hardware, or switching between Legacy BIOS and UEFI modes.
4. USB permissions
 - Allows you to enable or disable USB ports, control boot capability from USB drives, and manage legacy support for older devices
 5. Fan
 - adjusting fan speeds (RPM) relative to temperatures for optimal cooling and noise reduction
 6. Secure Boot
 - a UEFI firmware security feature that ensures only authorized, digitally signed software—such as trusted operating systems and bootloaders—can run during the startup process
 8. BIOS password
 - A BIOS password is a security measure that locks a computer's firmware to prevent unauthorized access or booting
 9. Temperature monitoring
 - the continuous or periodic tracking of thermal conditions using sensors, data loggers, and cloud-based systems to ensure safety, quality compliance, and operational efficiency
 10. Virtualization support
 - Run other operating systems within a single hardware platform
 - Virtualization added to the processor
 - Intel Virtualization Technology (VT)
 - AMD Virtualization (AMD-V)

▼ 3.5 TPM / HSM

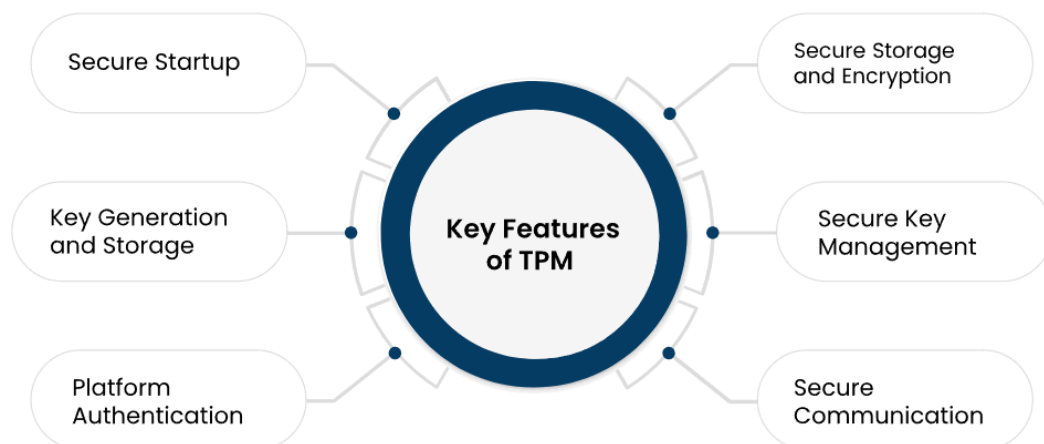


- TPM is for computer hardware and firmware security, while HSM is for cryptographic keys and other important data.
- When it comes to security, HSM is more secure than TPM.

▼ TPM - Trusted Platform Module

- Used on a single system
- Secure data on a local device
- Often built into a motherboard or available as an add-on module
- Function :

Key Features of TPM Security



▼ HSM - Hardware Security Module

- Used by many systems
- Secure data across multiple devices
- Often deployed as a high-end server or appliance in a data center
- Function :

Key Features of HSM Device



▼ 3.5 CPU architecture

▼ x86

1. Traditional 32-bit x86 instruction architecture
2. 32-bit OS cannot run 64-bit apps

▼ x64

1. 64-bit extension
2. also known as AMD64, x64, or Intel 64

▼ ARM - Advanced RISC [Reduced Instruction Set Computer] Machine

1. a power-efficient
2. RISC-based (Reduced Instruction Set Computer) design used in over 350 billion chips

▼ Core configurations

1. 2-4 Cores (Dual/Quad): Standard office work, basic web browsing
2. 6-8 Cores (Hexa/Octa): Gaming, content creation, and modern desktop tasks.
3. 10+ Cores: High-end workstations, video editing, and specialized development.

▼ 3.5 Cooling

▼ Fan

1. Active cooling components
2. Cool air is pulled through a personal computer



▼ Heat sink

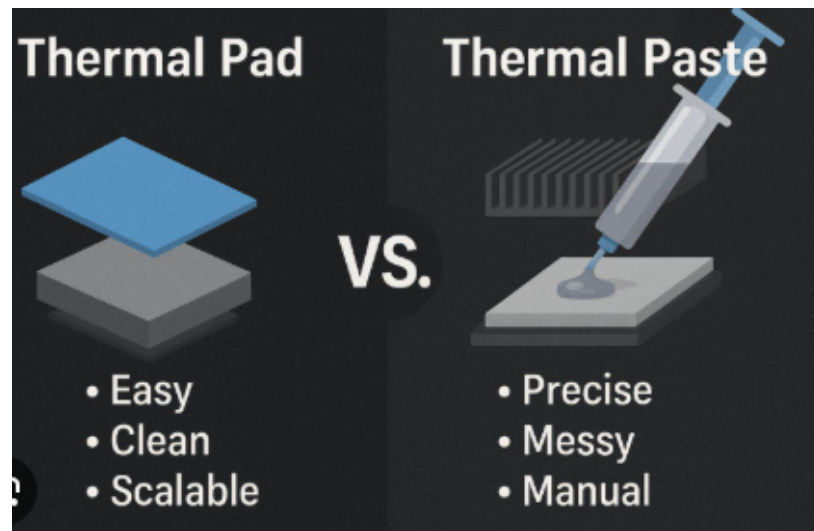
1. Passive cooling components
2. Transfers heat away from electronic to a cooling medium, typically air or liquid



▼ Thermal Paste/ Pad

1. Passive cooling components

2. Applied to fill the narrow gap between the CPU and heat sink for efficient heat transfer to prevent overheating



▼ Liquid-based cooling

1. Active cooling components
2. A high-efficiency thermal management method that uses a liquid medium to transfer heat



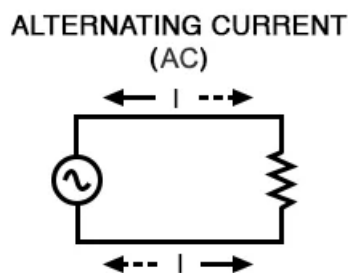
▼ 3.6 Power Supply



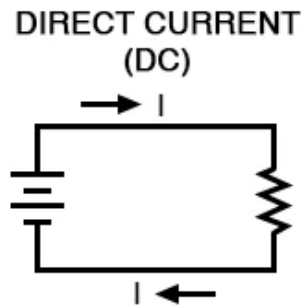
- Converts high-voltage alternating current (AC) from a wall outlet (110-240V) into stable, low-voltage direct current (DC) power (typically 3.3V, 5V, and 12V) required by computer components
- Watt = Volt * Amp ($60W = 120V * 0.5A$)
- Voltage (volt, V) = Electrical potential difference or pressure (Speed water)
- Ampere (amp, A) = The rate of electron flow (current) in a circuit. (Size hole from pipe)
- Power Supply measured in a unit known as: Watt

▼ DC / AC

- Alternating current (AC)
 - direction of current constantly reverses
 - Frequency of this cycle is important
 - power outlets use AC



- DC
 - Current moves in one direction with a constant voltage
 - PC components use DC, not AC



▼ Input 110–120 VAC vs. 220–240 VAC

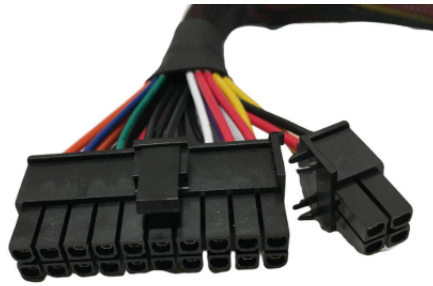
1. 110–120 VAC, 60 hertz (Hz) : the standard voltage range used for power outlets in United States
2. 220–240 VAC, 50 hertz (Hz) : the standard voltage range used for power outlets in Europe

▼ Output 3.3V vs. 5V vs. 12V

- +3.3V
 - NVMe M.2 slots
 - RAM slots
 - motherboard logic circuits
- +5V
 - USB ports and devices
 - Many components are now using +3.3 V
- +12V
 - Mechanical parts (e.g., case fans, motors, actuators)
 - Graphics cards
 - PCIe expansion cards

▼ 20+4 pin motherboard connector

1. Motherboards primarily use **12V, 5V, and 3.3V**
2. 20+4-pin connector to maintain compatibility with both 20-pin and 24-pin motherboards
3. the 4-pin section left unattached for 20-pin motherboards.



24-Pin (20+4)

▼ Redundant power supply

- Two (or more) power supplies
 - Internal to the server
- Each power supply can handle 100% of the load
 - Would normally run at 50% of the load
- Hot-swappable
 - Replace a faulty power supply without powering down



▼ Energy efficiency

- 80 PLUS certification
- PSUs based on their efficiency at various load levels
- Power supplies often range between 80% to 96%

- More efficiency means more DC power and less heat; save money!
 - PSU rating: White - Bronze - Silver - Gold - Platinum - Titanium (Less to High efficient)
-

▼ 3.7 Multifunction devices/printers (MFD/MFP)

▼ Properly unbox device and consider set-up location

- These can be large devices
 - Make sure you have enough room
 - Stay out of walkways
- Check the area
 - Power
 - Network
 - Accessibility

▼ PCL / PostScript

- PCL (Printer Command Language)
 - Generally more efficient for high-volume, everyday printing tasks than PostScript
 - Less suited for professional graphics or highly detailed printing tasks than PostScript
 - Less demanding on printer memory and processing power than PostScript
- PostScript
 - Slower printing performance compared to PCL due to complex processing
 - Better suited for professional design, desktop publishing, and graphic arts than PCL
 - More resource-intensive than PCL (requires more printer memory and processing power)
- Make sure the drivers match the printer

- PCL printer, PCL driver
- PostScript printer, PostScript driver

▼ Firmware

- software embedded within an MFD that provides basic instructions for its hardware to initialize and execute tasks

▼ Device connectivity

- USB - typically be used to set up a direct, wired connection between an MFD and a single computer host
- Ethernet - RJ45 connector
- Bluetooth - Limited range
- 802.11x (WIFI)
 - **Wi-Fi Infrastructure Mode:** Many devices using an access point
 - **Wi-Fi Ad hoc mode:** Direct link between wireless devices

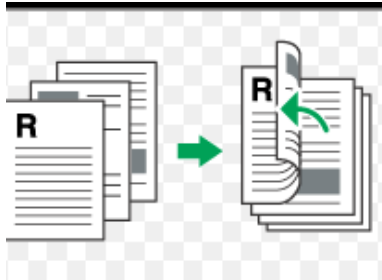
▼ Public/shared device

- Printer share disadvantage
 - Each client computer needs to install the correct printer drivers for the shared printer
 - Only available when the host computer is turned on and connected to the network
 - Can become a bottleneck, especially with many users or large print jobs
 - Offers limited control over user access, print quotas, and monitoring
 - If the host computer crashes or has issues, the printer becomes unavailable to everyone
- Print server
 - the highest availability of the network printing service
 - Print directly to the printer
 - Jobs are queued on the printer
 - Jobs are managed on the printer

- Web-based front-end
- Client utility

▼ Configuration settings

- Duplex
 - A printer's capability to print on both sides of a paper sheet



- Orientation
 - Portrait vs. Landscape, the default paper orientation is Portrait mode
 - The paper doesn't rotate
 - The printer compensates
- Tray settings
 - Printers can have multiple trays
 - Plain paper, letterhead, etc.
 - Choose the correct tray in the print dialog
 - allows configuration of a printout based on its paper size, type, or weight
- Quality
 - **Draft/Fast Mode:** Lowest quality, lowest DPI. Best for internal documents and saving ink.
 - **Normal/Standard Mode:** Ideal for everyday documents, balancing speed and quality.
 - **Best/High-Quality Mode:** High DPI, slow speed. Best for photos, presentations, or documents requiring high detail

- **Maximum DPI:** The highest possible resolution for specialized photo printing, often required for photographic paper.

▼ Security

- User authentication
 - Everyone can print
 - Set rights and permissions
 - Printing vs. managing the printer
- Badging
 - Authenticate when using the printer
 - Your job doesn't print until you use your employee badge
 - Quick and easy
- audit logs
 - Tracking of all printer activities, including user interactions, print jobs, and attempted access
- secured prints
 - PIN printing, requiring users to enter a numeric passcode before releasing print jobs

▼ Network scan services

- Email
 - scans are sent to your inbox
 - send a scanned document via an SMTP server
- SMB
 - Using SMB (Server Message Block)
 - scanned documents to be saved directly to shared folders on a local network
- Cloud services
 - scan a document from a multifunction printer and access it remotely from a personal online storage account
 - Scan to cloud (Google Drive, Dropbox, etc.)

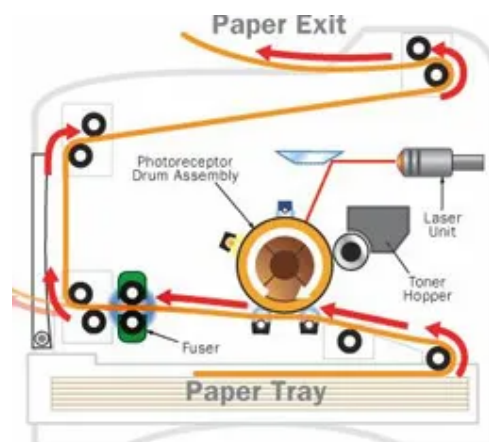
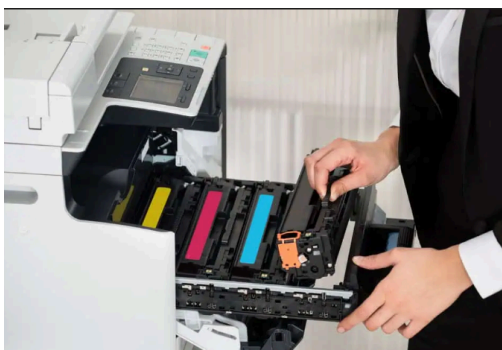
▼ Automatic document feeder (ADF)/flatbed scanner

- Automatic document feeder (ADF)
 - processing of multi-page documents
- flatbed scanner
 - a device that uses a flat glass surface on which documents, photos, or other materials are placed face down while a scanning mechanism moves underneath to capture the image



▼ 3.8 Printer Maintenance

▼ Laser Printer



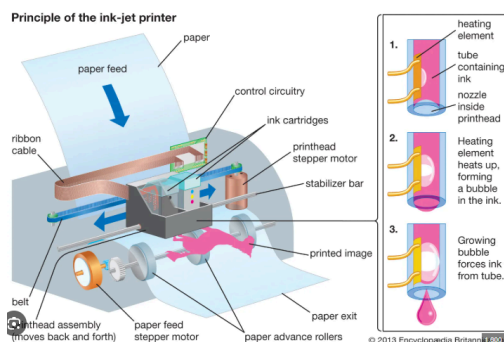
- Maintenance kit
 - A prepackaged set of laser printer spare parts

- Including components with a limited lifespan that must be replaced to maintain consistent performance and prevent failures
- installing a new toner cartridge
 - it is recommended to gently rock it back and forth several times in a horizontal motion
 - This step facilitates the uniform distribution of toner powder within the cartridge
 - which in turn contributes to a consistent print density throughout the lifespan of the cartridge.



- tools to clean laser printer
 - Toner vacuum
 - Toner cleaning wipes
 - Isopropyl alcohol

▼ Inkjet Printer



- Ink cartridge - the replaceable component used in the process of applying color to the page
- Printhead - an inkjet printer component that sprays microscopic droplets of ink through nozzles to form text or images on paper
 - ▼ methods for cleaning an inkjet printhead:
 - Running a built-in cleaning function from the printer settings menu
 - Wiping the nozzles with a lint-free cloth or swab dampened with distilled water or isopropyl alcohol
- Roller - A rubberized cylindrical component that grips and moves paper through the printing process in an inkjet printer
- Feeder - picking up individual sheets of paper from the input tray and feeding them into the print path
- inkjet calibration - To correct color printing errors and misalignments
- ▼ Paper jams issues
 - Obstruction in the paper path caused by foreign objects
 - Incorrect paper type or size
 - Paper tray overloading
 - Dirty, worn, or damaged paper feed rollers

▼ Thermal Printer



- Feed assembly - thermal printer's component responsible for moving paper through the printer during the printing process

▼ feed assembly maintenance procedure

- Inspect the feed assembly for worn-out or misaligned rollers
- Clean the rollers with a lint-free cloth and isopropyl alcohol to remove dust and debris
- If paper feeding issues persist, consider replacing the feed rollers

▼ Maintenance

- it is recommended to store thermal paper in a cool, dry place away from direct sunlight and heat sources
- The heating element, responsible for creating the image, should be periodically cleaned using a lint-free cloth or a specialized cleaning card lightly dampened with isopropyl alcohol to remove any residue buildup.
- Furthermore, it is important to regularly remove dust and debris from the rollers and paper feed areas by using compressed air or a soft brush to prevent paper jams and ensure smooth operation.

▼ Impact Printer (Dot matrix printer)



- Multipart - The printing paper designed to create duplicate copies during the printing process



- handling paper for an impact printer
 - Ensure perforated edges are properly positioned for continuous feeding
 - Align paper correctly in the tractor feed mechanism
-

4.0 Virtualization and Cloud Computing

▼ 4.1 Virtualization

▼ Purpose of virtual machines

- Sandbox Testing: Creating safe, isolated environments to test risky software or files without harming the host system.
- Test Development: Developers use VMs to create, deploy, and test software in various configurations or OS environments without needing dedicated hardware for each test, often utilizing snapshots to instantly revert to a prior state.
- Application Virtualization
 - Legacy Software/OS: Need to run different application versions on the same system
 - Cross-platform Virtualization: Enabling software to run on an unsupported operating system

▼ Requirements

1. Hardware

- CPU: Multi-core processor with virtualization extensions (Intel VT-x or AMD-V) enabled in BIOS/UEFI
- RAM: High capacity required to run multiple operating systems concurrently (typically 8GB minimum recommended).

- Storage: Fast storage (SSD) is crucial to handle virtual disk files for multiple VMs.

▼ Hypervisor Network

- shared networking
 - The VM receives its TCP/IP configuration settings from a virtual DHCP server managed by the hypervisor
 - The VM uses the IP address of the host machine for outbound communication via NAT
 - The VM's private address is not advertised on the LAN preventing direct inbound connections
- bridged networking
 - The VM's network adapter is connected directly to the physical network through the host's network interface
 - The VM can communicate directly with all other devices on the physical network and the Internet, just like any other physical machine
 - The VM can communicate directly with all other devices on the physical network and the Internet, just like any other physical machine

▼ Hypervisor Security

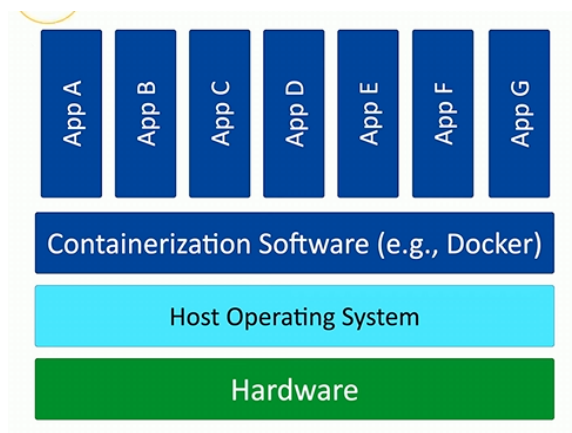
- Hypervisor is a sweet spot for the bad guys
 - No significant vulnerabilities yet
- VM escaping
 - Malware recognizes it's on a virtual machine
 - Malware compromises the hypervisor
 - Malware jumps from one guest Os to another
- Many hosted services are virtual environments
 - Malware on one customer's server can gather information from another

▼ Desktop virtualization

- Desktop virtualization infrastructures (DVI)
 - Applications actually run on a remote server
 - Local device is a keyboard, mouse, and screen
 - Desktop as a Service (DaaS)

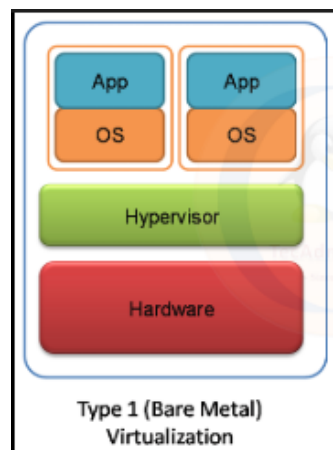
▼ Containers

a lightweight, OS-level technology that packages applications and their dependencies together, allowing them to run in isolated environments on a shared host OS kernel

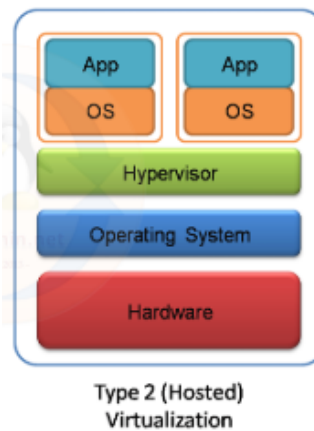


▼ Hypervisors

- Type 1: Bare Metal
 - The hypervisor is the primary operating system



- Type 2: Hosted
 - Hypervisor runs in the existing OS



▼ 4.2 Cloud Computing

Common Cloud Models:

▼ Private cloud

- A cloud infrastructure that is exclusively used by a single organization
- Organizations with strict compliance and security requirements would typically use a private cloud

▼ Public cloud

- Available to everyone over the Internet
- shared across multiple organizations or tenants

▼ Hybrid cloud

- A mix of public and private
- Combines the benefits of private and public cloud deployment models

▼ Community cloud

- Several organizations share the same resources
- A cloud environment developed for universities conducting joint research projects
- A shared cloud used by several government agencies
- A cloud-based solution for public safety organizations with similar data handling policies

LaaS, PaaS, SaaS:

▼ Infrastructure as a service (LaaS)

- Sometimes called Hardware as a Service (HaaS)
- A cloud computing service model where clients, instead of buying hardware and software, access computing infrastructure as an outsourced service from suppliers who own and maintain all the necessary equipment and software

▼ Platform as as service (PaaS)

- Gives users the benefits of using the cloud and the ability to customize their own development
- A cloud computing service model would provide the best solution for a web developer intending to create a web app

▼ Software as a service (SaaS)

- Allows end users to access their email, documents, and spreadsheets directly within their web browser
- A cloud computing service model that provides users with remote access to applications over the Internet on a subscription

Cloud characteristics:

▼ Shared resources vs Dedicated resources

- Shared resources:
 - Refer to an arrangement where multiple customers use the same underlying physical infrastructure
 - Lower costs due to economies of scale
 - Possible performance degradation during high demand periods
- Dedicated resources:
 - Refer to resources that are reserved for a single customer
 - Consistent and reliable performance
 - Higher, fixed costs regardless of resource use
 - Greater control and stronger data isolation

- refer to private cloud

▼ Metered utilization

- a billing model where customers are charged based on actual resource consumption.
 - Ingress: Cost to data entering the cloud
 - egress: Cost to data leaving cloud

▼ Elasticity

- allocation adjusts automatically based on demand
- Allows resources to scale up or down dynamically based on demand
- minimizes the risk of having too many idle resources (over-provisioning) or too few resources during peak times (under-provisioning), leading to efficient resource management.

▼ Availability

- Systems are always available
- Redundancy provides availability
- Ensuring services are accessible with minimal interruption
- **Service Level Agreement (SLA):** 99.9 Uptime percentage that services are guaranteed to be operational.

▼ File synchronization

- Information can be duplicated across cloud locations
- Reflecting file edits made on one device across all other connected devices

▼ Multitenancy

- Many different clients are using the same cloud infrastructure and applications completely separate
 - Enables several customers to share a single software deployment without accessing each other's data
-

5.0 Hardware and Network Troubleshooting

▼ 6 Step Troubleshooting Methodology

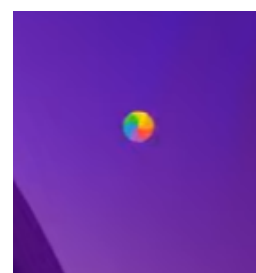
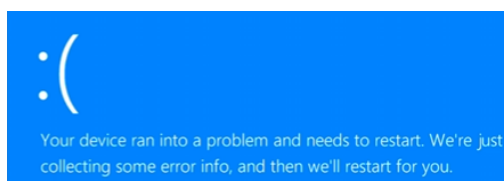
1. Identify the problem
2. Establish a theory of probable cause (question the obvious)
3. Test the theory to determine the cause
4. Establish a plan of action to resolve the problem & implement the solution
5. Verify full system functionality and, if applicable, implement preventative measures
6. Document the findings, actions, & outcomes

▼ 5.1 Hardware Troubleshooting

▼ POST - Power On Self Test

- Beeps are diagnostic signals from your BIOS
- No Beep: No power, dead motherboard, or failed CPU
- 1 Beep: System OK (depends on BIOS)
- Repeating Beeps: RAM, GPU, or CPU issue
- Fixes: Check BIOS beep codes, reseal RAM/GPU, clear CMOS, test parts individually
- Tip: Always test on a non-conductive surface

▼ Proprietary crash screens (BSOD/ Mac Pinwheel)



- Indicate system or driver failures

- Causes: Bad RAM, overheating CPU, driver/hardware conflict, motherboard fault
- Fixes: Run memory diagnostics, monitor temps, check Event Viewer, update BIOS/drivers

▼ Blank Screen (No Display)

- Causes: GPU/ RAM not seated, CPU/motherboard fault, no power
- Fixes: Test monitor, reseal components, test one RAM stick, inspect CPU socket



▼ No Power

- Causes: Bad PSU, loose cable, dead Motherboard, damaged power button, wrong voltage
- Fixes: Test PSU, check connections, ensure correct voltage, never open a PSU

▼ Sluggish Performance (Lag)

- Causes: Failing RAM, overheating, dust, malware, outdated firmware
- Fixes: Use Task Manager, clean dust, reapply thermal paste, run malware scans, upgrade hardware

▼ Overheating

- Causes: Bad cooling, dried thermal paste, poor airflow, overclocking
- Fixes: Clean fans, reapply paste, add case fans, monitor temps

▼ Burning Smell

- Causes: Failing PSU, fried capacitor, overheated CPU, dust short
- Fixes: Power off immediately, inspect components, replace PSU / Motherboard

▼ Random Shutdowns

- Causes: Overheating, power surges, RAM/ PSU failure, BIOS settings
- Fixes: Monitor temps, use surge protector, test RAM, update BIOS

▼ Application Crashes

- Causes: Faulty RAM / CPU, software conflict, malware ,overclocking
- Fixes: Run sfc /scannow, test RAM, disable overclocking, check reliability monitor

▼ Unusual Noises

- Causes: Fan bearings, loose cables, coil whine, shorting components
- Fixes: Clean / secure cables, replace fans, isolate sound using a cardboard tube

▼ Capacitor Swelling

- Causes: degraded performance, such as system instability, random freezes, or sluggishness, without immediately triggering errors or shutdowns.
- Fix: Replace the motherboard (repairs usually not worth it)



▼ Inaccurate System Date / Time

- Causes: Dead CMOS battery

- Fix: Replace CR2032 battery, reset BIOS date / time, save & exit



▼ 5.2 Troubleshoot drive and RAID

▼ RAID LED Status Indicators

- Green: Drive operating normally; data safe
- Amber / Yellow: Warning or minor issue (slow reads /writes)
- Red: Critical failure; possible failed drive or RAID disk error
- Tip: Always verify LED meanings in the manufacturer's manual – codes can vary by model

▼ Grinding Noises

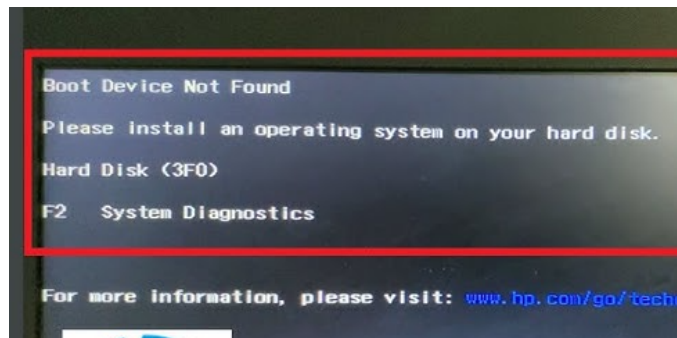
- Indicates wear or misalignment of moving parts (read/ write heads, spindle motor)
- Stop using the drive immediately & backup data if possible
- Continued use can cause irreversible data loss

▼ Clicking Sounds

- Signals the actuator arm or heads can't locate data sectors
- Implies a near-total mechanical failure; backup & replace immediately

▼ Bootable Device Not Found

- Common Causes:
 - Corrupted OS on boot drive
 - Failed RAID member or disk
 - BIOS/UEFI misconfiguration pointing to wrong drive
- Action: Verify BIOS settings & RAID integrity



▼ Data Loss | Corruption

- Causes: Power loss, RAID sync failure, or physical disk issues
- Symptoms:
 - Inaccessible or unreadable files
 - Corrupted documents or images
 - Error messages like "Drive not found" or "Data integrity check failed"
- Response: Use recovery tools & attempt RAID rebuilds to minimize downtime

▼ RAID Failures

- Single Drive Failure (RAID 1): Replace the failed drive immediately to restore redundancy
- Multiple Drive Failure (RAID 5 / 6): Two failed drives in RAID 5 or three in RAID 6 may cause total data loss
- Actions: Check LED indicators, listen for noises, & attempt a controlled rebuild

▼ Self-monitoring and Reporting Technology (S.M.A.R.T.) failure

- High temperatures, read/write errors, & bad sectors suggest imminent drive failure
- Fix: Backup data immediately, replace the drive, & run diagnostic utilities

▼ Extended Read / Write Times

- May signal bad sectors, degraded RAID arrays, or an overworked controller

- Fix: Use disk management & S.M.A.R.T. tools to isolate & replace slow drives
- ▼ Low performance input/output operations per second (IOPS)
 - Caused by degraded disks, file fragmentation, or network bottlenecks
 - Fix :Check RAID health, defragment disks, & upgrade old drives or controllers
- ▼ Missing Drives in OS
 - Often due to loose cables, bad sectors, or misconfigured arrays.
 - Fix: Verify physical connections, RAID setup, & perform OS-level disk checks
- ▼ Array missing
 - usually indicates drive failures, controller issues, or improper shutdown, often rendering data inaccessible
 - Fix: Do not reinitialize or format the drives. Instead, re-seat drives, check cable connections, check BIOS/controller status
- ▼ Audible alarms
 - a security feature designed to alert administrators of a hardware fault.

▼ 5.3 Display Troubleshooting

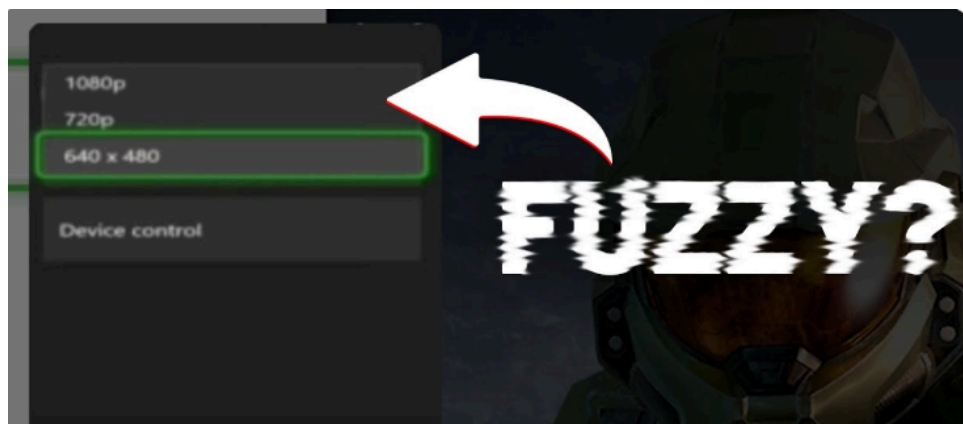
- ▼ Incorrect input source
 - Symptoms: Blank screen, "No Signal", wrong display output
 - Fix: Verify correct input (HDMI / VGA /DisplayPort), test source device, toggle laptop display mode (Windows + P)
- ▼ Physical Cabling Issues
 - Symptoms: Flickering, distorted or no image
 - Fix: Reseat or replace damaged cables; test alternate ports or cables
- ▼ Burnt-Out Projector Bulb
 - Symptoms: Dim image, bulb warning, or no projection

- Fix: Check bulb indicator, replace lamp, inspect cooling system



▼ Fuzzy Image

- Symptoms: Blurry text or pixelated display
- Fix: Match display's native resolution, use quality cables, & test another monitor



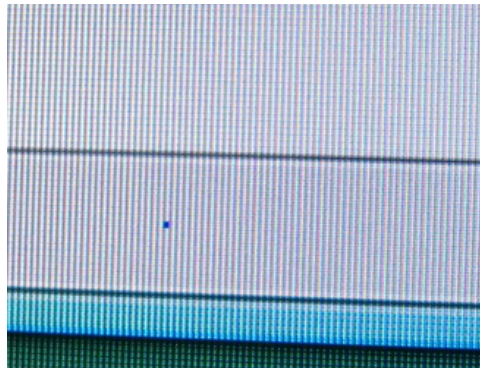
▼ Display Burn-In

- Symptoms: Ghost images or static shadows
- Fix: Enable screen savers / pixel shifting, lower brightness, or use burn-in repair tools

Departures - Terminal 1						
DESTINATION	AIRLINE	FLIGHT	TERMINAL	GATE	TIME	REMARKS
J. Cabo	American Airlines	1225	D	D31	9:50a	On Time
Sacramento	American Airlines	1103	C	C33	9:35a	Now 9:50a
San Jose	American Airlines	1103	C	C33	9:35a	On Time
San Jose	Delta Air Lines	1096	E	E13	8:30a	Now 9:47a
San Jose	American Airlines	609	C	C5	9:25a	On Time
San Jose	American Airlines	3417	B	B24	8:55a	On Time
San Jose	American Airlines	3485	B	B24	10:20a	On Time
San Jose	American Airlines	1839	A	A34	8:15a	Now 8:40a
San Jose	American Airlines	1978	D	D17	9:25a	On Time
San Jose	American Airlines	1789	D	D16	8:40a	On Time
San Jose	United Airlines	1247	B	B29	7:25a	On Time
San Jose	United Airlines	3062	A	A23	7:55a	On Time
San Jose	United Airlines	5865	C	C11	8:55a	On Time
San Jose	Alaska Airlines	1801	A	A24	10:00a	On Time
San Jose	American Airlines	9235	D	D20	10:00a	On Time
San Jose	American Airlines	8352	D	D29	7:55a	On Time
San Jose	TAM	1347	A	A20	10:00a	On Time
San Jose	American Airlines	2183	D	D38	10:15a	On Time
San Jose	American Airlines	2058	D	D29	8:15a	On Time
San Jose	Delta Air Lines	9011	E	E10	7:00a	On Time

▼ Dead Pixels

- Symptoms: Stuck black or bright dots
- Fix: Try pixel-repair software; contact manufacturer for warranty replacement



▼ Flashing / Flickering Screen

- Symptoms: Intermittent blackouts or flickers
- Fix: Check cable integrity, update graphics drivers, & adjust refresh rate

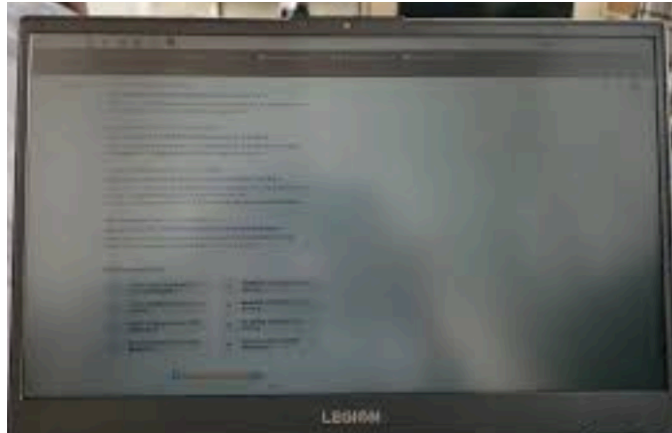
▼ Incorrect Color Display

- Symptoms: Washed-out or tinted visuals
- Fix: Calibrate colors via monitor OSD or graphics settings; test another display

▼ Dim Image

In older laptops with CCFL backlights, the inverter powers the light; if it fails, the screen will appear extremely dim while the image remains.

- Symptoms: Screen visibility too low even at max brightness
- Fix: Adjust brightness / backlight settings; test alternate power or display, replace inverter

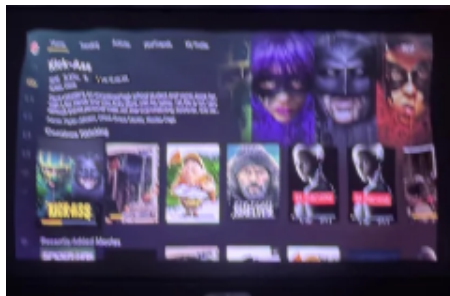


▼ Intermittent Projector Shutdown

- Symptoms: Turns off after short use
- Fix: Clean vents, check fans, ensure proper Ventilation, & replace aging lamps

▼ Sizing / Distortion Problems

- Symptoms: Stretched, cropped, or garbled image with misaligned colors and jagged text
- Fix: Adjust aspect ratio, resolution, Reseat or replace the video cable
-



▼ Intermittent Projector Shutdown

- Symptoms: Turns off after short use
- Fix: Clean vents, check fans, ensure proper ventilation, & check the cooling system

▼ 5.4 Mobile device Troubleshooting

▼ Poor battery health

- Common Signs:
 - Battery drains unusually fast
 - Phone shuts off randomly even with charge remaining
 - Battery percentage fluctuates erratically
- Underlying Cause:
 - Every charge cycle slightly reduces battery's capacity
 - Over time, battery struggles to hold a full charge efficiently
 - Heat, overcharging, & heavy app usage accelerate wear
- How to Fix or Improve
 - Check Battery Health
 - Limit background apps & processes
 - Turn off GPS/ location when not needed
 - Reduce screen brightness or enable battery saver mode
 - Replace the battery if capacity is severely degraded (especially in older devices)

▼ Swollen Battery



- Visual Warning Signs
 - Phone case no longer fits or pops open unexpectedly
 - Screen bulging, lifting, or separating from the frame
 - Device feels unusually warm even when idle
- What's Happening Inside
 - Lithium-ion battery is breaking down internally, generating gas build up

- This causes battery to expand, putting pressure on screen & casing
- Often triggered by age, overheating, or overcharging
- What to Do Immediately
 - Stop using the device right away
 - DO'NOT puncture, press, or attempt to remove the swollen battery yourself
 - Power off phone & place it in a non-flammable area if possible
- Safe Next Steps
 - Take device to a certified repair center for professional removal
 - Dispose of properly at an approved e-waste or recycling facility
 - Avoid throwing battery in household trash or regular recycling bins

▼ Broken Screen

- Common Signs of a Broken Screen
 - Cracked or shattered glass
 - Unresponsive or erratic touch input
 - Visible lines, flickering, or dead pixels on the display
- How to Fix It
 - Assess the Damage: Determine whether it's only cosmetic (glass) or if the digitizer or LCD is damaged too
 - Prevent Further Damage: Apply a temporary screen protector or clear tape to keep cracks from spreading
 - Replace the Screen

▼ Improper Charging

- Signs of Improper Charging
 - Device charges slowly or not at all
 - Intermittent charging - starts & stops randomly

- Only charges with a specific cable or when held at a certain angle
- Possible Causes
 - Faulty charging port (loose or damaged pins)
 - Damaged charging cable (frayed or bent connectors)
 - Software glitch preventing proper power draw
 - Power source issue (faulty outlet, adapter, or USB port)
- How to Fix
 - Test with a different charger & power outlet
 - Clean the charging port using a soft brush or compressed air
 - Inspect the port for visible debris or bent contacts
 - Restart the device to clear minor software issues
 - Replace the charging port if physically damaged (common in high-use phones)

▼ Poor / No Connectivity

- Symptoms
 - WiFi fails to connect or repeatedly disconnects
 - Bluetooth devices won't repair or maintain connection
 - Mobile data unavailable or extremely slow
 - Web pages & apps stuck "loading" with spinning wheel
- Troubleshooting Steps
 - Toggle Airplane Mode: Turn it on, wait 10 seconds, then turn it off to reset connections
 - Restart Device: Clears temporary glitches & refreshes network adapters
 - Forget & Reconnect to Network: Removes corrupted connection profiles
 - Reset Network Settings: Restores default WiFi, Bluetooth, & cellular settings

- Update Firmware: Fixes bugs & improves connectivity performance
- Bluetooth Fix: Unpair, restart both devices, then re-pair
- Additional Tip
 - If multiple devices on same network experience issues, Problem may lie with the router or network provider

▼ Liquid Damage

- Signs
 - Device won't appear on or randomly shuts down
 - Screen flickering, dim display, or discoloration
 - Distorted or no audio from speakers / headphones
 - Corrosion or residue visible in charging port or SIM tray
- Steps to Fix
 - Power off the device immediately to prevent short circuits
 - Remove accessories, SIM card, & memory card
 - Place device in a dry, warm area with good airflow
 - Use silica gel packets (more effective than rice)
 - Avoid heat guns, ovens, or blow dryers- can cause further damage
- the best practices for drying a smartphone
 - Apply desiccant-based drying methods to remove moisture
 - Keep the device turned off for at least 24–48 hours before testing
 - Place the device in a sealed bag with silica gel packets

▼ Overheating

- Signs of Overheating
 - Device feels unusually hot to the touch
 - Random shutdowns or restarts
 - Frequent app crashes

- On-screen overheating or temperature warnings
- Common Causes
 - Running resource-heavy apps (e.g., gaming, 4Kstreaming)
 - Direct sunlight exposure for extended periods
 - Poor ventilation (e.g., using under a blanket or pillow)
 - Malware or rogue apps consuming system resources
- How to Fix It
 - Close unused background apps
 - Limit intensive use in hot environments
 - Remove the phone case to improve airflow
 - Restart the device to reset processes
 - Run malware scans & remove suspicious apps
 - Using an original or certified device charger

▼ Digitizer Issues

- Common Signs of Digitizer Problems
 - Touch inputs not registering at all
 - Only part of the screen responds to touch
 - Phantom touches or ghost typing occur unexpectedly
- Troubleshooting & Fixes
 - Restart the Device: Clears temporary software glitches
 - Clean the Screen: Dirt, oil, or moisture can interfere with sensitivity
 - Remove Screen Protector: Some protectors block or distort touch signals
 - Boot Into Safe Mode: Helps rule out malfunctioning apps
 - Factory Reset (Last Resort): Eliminates software corruption
 - Hardware Replacement: If issues persist, the digitizer itself may need professional repair or replacement

- Reduce the risk of digitizer issues
 - no drop phone, install screen protector, keep the phone far away from water and heat.

▼ Physically Damaged Ports

- Common Signs
 - Plug doesn't stay securely inserted
 - No sound through headphones or only one side works
 - Device won't charge or requires wiggling the cable
- Troubleshooting Steps
 - Inspect for debris or lint using a flashlight
 - Gently clean with a wooden toothpick or soft brush
 - Test with a different cable or charger
 - Examine for bent or broken pins inside the port
- Repair or Replacement
 - Minor debris removal can restore functionality
 - If internal damage or broken pins are present, port replacement is typically required
 - Avoid using excessive force or metal tools during cleaning to prevent further damage

▼ Malware

- Signs of Infection
 - Constant pop-ups or ads appearing out of nowhere
 - Noticeable slowdown in performance
 - Unknown or unwanted apps installing themselves
 - Battery draining unusually fast
- How to Fix It
 - Boot device into Safe Mode (if supported) to isolate the issue
 - Uninstall suspicious or unknown apps

- Run a trusted mobile antivirus scan
- Clear cache & storage to remove residual files
- Factory reset if the infection is deep or persistent

▼ Cursor Drift / Touch Calibration Issues

- Signs of the Problem
 - Inaccurate touch response or delayed input
 - Phantom movements or random screen actions
 - Difficulty typing or selecting small icons
- How to Fix It
 - Reboot the device to clear temporary glitches
 - Remove screen protector- may interfere with touch sensitivity
 - Use built-in calibration tools(available on some Android devices)
 - Update the operating system for bug &driver fixes
 - Factory reset as a last resort if all else fails

▼ Unable to Install New Applications

- Check available storage space
- MDM - Policy enforcement

▼ Stylus Does Not Work

- Signs of the Issue:
 - device's stylus functions intermittently
 - touchscreen responds normally to finger input but stylus not function
- Troubleshooting Steps:
 - Inspect the stylus tip for damage or wear
 - Check the battery inside the stylus (if it uses one)
 - Check for firmware or software updates

▼ Degraded Performance

- Lag or unresponsive interface

- Apps frequently crashing or freezing
- Troubleshooting Steps
 - Restart the device
 - Clear the cache partition
 - Close or uninstall resource-heavy or unused apps
 - Free up storage by deleting unnecessary files or apps
 - Disable animations via developer options for smoother performance
 - Update the OS to ensure optimal compatibility & efficiency

▼ 5.5 Network Troubleshooting

▼ Intermittent Wireless Connectivity

| Internet auto disconnect

- Common Signs:
 - Weak WiFi due to distance, walls, or obstacles
 - Wireless channel interference from nearby networks or devices (microwaves, cordless phones, Bluetooth)
 - Router firmware issues leading to unstable performance
 - Outdated or corrupt network drivers on the client device
 - Roaming behavior when devices switch between multiple access points
- Troubleshooting Steps
 - Check Signal Strength: Move closer to router; if stability improves, signal is the issue
 - Scan for Interference: Use a WiFi analyzer tool to find congested channels / Switch router to channels 1, 6, or 11 on 2.4 GHz for better performance
 - Update Drivers & Firmware
 - Adjust Roaming Settings
- Add on

- tool troubleshooting intermittent wireless connectivity: Wi-Fi analyzer



▼ Slow Network Speeds

- Common Signs:
 - Bandwidth Hogs: Other device take the network
 - Network Congestion: Peak-hours slowdown
 - Outdated Hardware: Old routers
 - Malware / Background Processes: Hidden apps
 - ISP Issues: internet provider give poor line
- How to Troubleshoot
 - Run speed test
 - Check task manager network
 - Restart router
 - Upgrade router
 - Configure QoS on the router to focus VoIP or video call

▼ Limited Connectivity

connected to a local router but have no access to the internet
the computer cannot reach a DHCP server to obtain a valid IP address

- Why
 - IP address conflict: 2 device use same IP
 - DHCP issues: Router no assign valid IP
 - Gateway problem: Default gateway unreachable

- DNS miss configuration: name won't resolve
- Physical issues: Cable damaged or unplugged
- Troubleshooting
 1. Release & Renew IP address: Command Prompt → ipconfig /release → ipconfig /renew
 2. Check Static IP Settings: Ensure device isn't set to a static IP outside DHCP range
 3. Verify Gateway & DNS: Use ipconfig /all to confirm correct network
 4. Swap Physical Components: Try a different Ethernet cable or router port for wired connections

▼ Jitter

| Internet not stable

- Why
 - Network Congestion
 - Buffer Bloat
 - Poor ISP Routing
 - Inadequate QoS Settings
- Troubleshooting
 - Run a Latency test
 - Try a Wired Connection
 - Enable and adjust QoS
 - Stop large download and streaming.

▼ Poor VoIP Quality

- Why:
 - High latency affecting real-time communication
 - Jitter causing inconsistent packet delivery
 - Packet loss interrupting audio streams

- Firewall / NAT configuration blocking or mangling VoIP traffic
- Misconfigured or incompatible audio checks
- Troubleshooting Steps
 - Test Your Internet Connection

▼ Port Flapping

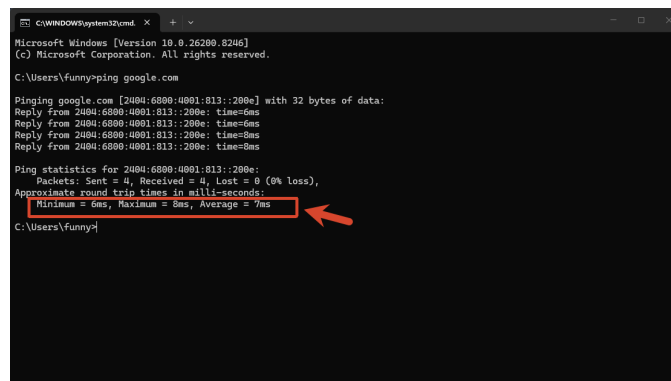
a network issue where a physical interface (switch port, NIC) rapidly toggles between up and down states

- Why
 - Faulty Switch or Port
 - Bad Ethernet Cable
 - Power-Saving Features
 - Duplex Mismatch
- Troubleshooting Steps
 - Move the cable to difference switch port
 - Replace with good cable
 - Disable power saving
 - Match duplex setting

▼ High Latency

- **a delay between sending data and receiving a response, causing lag, slow loading, and poor real-time performance (gaming/video calls)**
- Why
 - Network congestion due to excessive traffic
 - Physical distance between the sender and receiver of data
 - Wireless interference and signal degradation
 - Outdated or inefficient network hardware
- Troubleshooting
 - ▼ 1. ping check : cmd > ping google.com

- **0–30 ms: Excellent.** Ideal for competitive gaming and high-quality video calls.
- **30–70 ms: Good.** Very smooth for casual gaming and streaming.
- **70–100 ms: Average.** Noticeable in fast-paced games but acceptable for web browsing.
- **100–150 ms: Fair.** You may experience "lag" or stuttering in real-time applications.
- **150ms+: Poor.** Significant delays; likely to cause disconnection or heavy buffering.



```

C:\WINDOWS\system32\cmd
Microsoft Windows [Version 10.0.26299.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\funny>ping google.com

Pinging google.com [2404:6800:4001:813::200e] with 32 bytes of data:
Reply from 2404:6800:4001:813::200e: time=6ms
Reply from 2404:6800:4001:813::200e: time=6ms
Reply from 2404:6800:4001:813::200e: time=8ms
Reply from 2404:6800:4001:813::200e: time=8ms

Ping statistics for 2404:6800:4001:813::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
    Minimum = 6ms, Maximum = 8ms, Average = 7ms
C:\Users\funny>
  
```

2. Test at different times of day to identify peak congestion or ISP throttling patterns
3. **Switch to LAN**
4. contact ISP

▼ External interference

- Why
 - radio frequency (RF) noise from non-controlled, outside sources (e.g., faulty equipment, neighboring networks) that degrades performance, causes dropped calls, and reduces throughput.
 - Interference reduces signal quality, causing drops, slow speeds, or jitter
- Troubleshooting
 - Use 5GHz WiFi
 - Relocate the Router

- Switch Channel
- Use LAN

▼ Authentication Failures

- Why
 - Wrong WiFi password (most common cause- even one character off breaks the connection)
 - Credentials recently changed (router password updated, enterprise login rotated)
 - Expired user password, Disabled or deleted user account. User account locked out, Time synchronization issues
- Troubleshooting
 - Double-check the password
 - Forget & reconnect to the network

▼ Intermittent Internet Connectivity (LAN)

- Why
 - Loose / Damaged Cables
 - ISP Issues
 - Router Overheating
 - DHCP Lease Problems
- Troubleshooting
 - Inspect Physical Connections
 - Review Modem / Router Logs
 - Test Additional Devices
 - Power cycle (restart) modem and router
 - Contact ISP

▼ 5.6 Printer Troubleshooting

▼ Lines down the printed pages

- Why

- Dirty printhead or roller
- Damage, leaking, or low toner cartridge
- Imaging drum worn or failing (laser printer)
- Troubleshooting
 - Run the printer's built in cleaning cycle (via setting)
 - manually inspect and gentle clean print head (inkjet printer)
 - Replace toner or drum unit if cleaning doesn't resolve steels (Laser printer)

▼ Garbled Print

- Why
 - Outdate, incorrect printer driver install
 - Corrupted print job stuck
 - Printer language mismatch (such as PCL vs. PostScript)
 - incorrect printer configurations
 - damaged interface cables
 - formatting issues at the application level
- Troubleshooting
 - Update, reinstall, or replace the printer driver
 - Clear the printer queue
 - Verify printer language settings in printer properties

▼ Paper Jams

- Why
 - Paper loaded incorrectly
 - Curled or deformed paper sheets
 - Incorrect paper size or type
 - High humidity levels affecting paper
 - Overloaded paper tray
 - Dirty or worn paper pickup rollers

- Troubleshooting
 - Power off printer
 - Open all access panel and slowly remove jammed paper
 - Check for worn or dirty paper pickup rollers
 - Reload the tray using high quality paper
 - Replace rollers if show signs of wear or slipping

▼ Faded Prints

- Why
 - Low toner / ink density
 - Dried or evaporated ink, Print head misalignment (Inkjet printer)
 - Toner not fusing properly (Laser printer)
- Troubleshooting
 - Open printer setting increase print quality
 - Clean and replace print head (Inkjet printer)
 - Swap in a fresh toner (Laser printer) or ink (Inkjet printer) cartridge.
 - inspect the fuser unit for malfunction (Laser printer)

▼ Paper Not Feeding

- Why
 - Worn-out or dusty feed rollers
 - Foreign objects or paper debris in the paper path
 - Overloaded paper tray
 - Paper quality or type issues
- Troubleshooting
 - Clean feed rollers
 - Reload & align paper
 - Confirm correct paper type
 - Ensure tray is fully seated

▼ Multipage Misfeed

- Why
 - Paper quality issues
 - Worn or faulty separation pad
 - Overfilled paper tray
 - Worn or dirty pickup rollers
 - Humidity affecting paper
- Troubleshooting
 - Use correct paper type
 - avoid humidity
 - replace separation pads if problem persists
 - Replace worn feed rollers

▼ Multiple Prints Pending in Queue

- Why
 - Frozen or jammed print queue preventing jobs from processing properly
 - Printer goes offline, causing jobs to pile up behind the scenes
 - Print Spooler service crash on Windows, interrupting job management
 - Network delays or connectivity issues that slow job delivery
- Troubleshooting
 - **Cancel All Jobs**
 - Restart the Print Spooler service Windows Path: `services.msc` → `Print Spooler` → `Restart`
 - Reboot printer and computer
 - reset printer if still stuck

▼ Speckling on printed pages

- Why

- Leaking toner cartridge (Laser Printer)
- Paper dust, debris, or excess toner buildup (Laser Printer)
- Contaminated or faulty imaging drum (Laser Printer)
- Excess paper dust, debris, or ink residue on the paper path (Inkjet Printer)
- Excess paper dust, debris, or ink residue on the paper path (Inkjet Printer)
- Leaking ink cartridge(Inkjet Printer)
- Troubleshooting
 - Remove the toner cartridge & gently shake it to redistribute toner evenly
 - Clean the interior of the printer using a soft, lint-free cloth while the device is powered off
 - Replace defective cartridges or components is shaking / cleaning doesn't resolve the issue

▼ Double/ echo images on the print

- Why
 - residual latent image on the imaging drum being reused during the same print cycle
 - Fuser Unit Malfunction: Not heating evenly or failing to properly bond toner to paper
- Troubleshooting
 - checking the drum for wear or residual toner buildup
 - verifying that the fuser is operating at the correct temperature and effectively bonding the toner to the paper
 - replacing any faulty components as needed.

▼ Grinding noise

- Why

Inkjet Printer

Laser Printer

- Worn, damaged or misaligned paper feed rollers
- Small paper pieces or foreign objects stuck in the paper path
- Printhead carriage mechanism malfunctions (e.g., misaligned or obstructed carriage)
- Faulty or worn fuser rollers and bearings
- Worn or damaged gears and bearings affecting the imaging drum assembly
- Small foreign objects or paper pieces stuck in the paper path
- Improperly aligned, worn, or faulty paper feed rollers
- Troubleshooting
 - Power off & unplug the printer before inspecting
 - Open all accessible panels & trays to visually inspect the interior
 - Gently remove any debris, jammed paper, or obstructions
 - Check roller alignment & reseal rollers if needed
 - If grinding continues, internal gear or rollers likely need professional servicing

▼ Finishing issues

- Refers to post-printing problems such as stapling documents and punching holes in printouts
- locating the hole-punch waste container on the printer and emptying its contents
- Adjusting the placement of staples and holes on the printed document can be performed via the printer's software.

▼ Incorrect page orientation

- Why
 - Incorrect orientation
- Troubleshooting
 - Review the page setup in the specific
 - Confirm & correct orientation in the printer driver settings

- Use "Print Preview" to verify layout
- Perform a small test print
- ▼ Tray not recognized
 - Why
 - A dirty or malfunctioning paper tray sensor
 - Printer settings that do not match the paper type or size
 - Unseated or improperly inserted paper tray
 - Troubleshooting
 - **Reseat and Inspect the Tray**
 - **Clean the Sensors and Rollers**
 - **Verify Printer Settings**
 - **Perform a Hard Reset**
- ▼ Connectivity issues
 - Why
 - USB Cable Unplugged / Loose
 - WiFi Printer Disconnected
 - Printer on a Different Subnet
 - IP Address Conflict / Change - DHCP assign new address
 - Bluetooth Printer Not Paired
 - Troubleshooting
 - **Check Physical/Network Connection:** Ensure USB cables are secure or that the printer is connected to the correct Wi-Fi/Ethernet network.
 - **Verify IP Configuration:** Confirm the printer has a valid IP address, subnet mask, and default gateway (often printed via a configuration page).
 - **Restart Devices:** Power cycle the printer, computer, and network router/switch to clear temporary glitches.

- **Check Link Lights:** Observe the network port lights on the printer for active link and data activity.
- **Update/Reinstall Drivers:** Ensure the latest driver is installed on the computer to avoid communication failures.
- **Check Print Spooler:** Restart the Print Spooler service on Windows to clear hung print jobs.
- **Check Firewall/Security Settings:** Confirm that firewall rules allow traffic to the printer's IP address.
- **PING the Printer:** Use the `ping [IP address]` command from the workstation to test connectivity.

▼ Frozen print queue

- Why
 - Corrupted print job blocking the queue
 - Print Spooler service crash disrupting print processing
 - Excessive queued jobs
- Troubleshooting
 - Manually cancel all pending jobs in the printer settings
 - Restart the Print Spooler service
 - Delete temporary spool files - Windows Path:

`C:\Window\System32\spool\PRINTERS`
 - Power cycle the computer and/or printer
 - Update or reinstall the printer driver/software

Issue	Likely Cause	Quick Fix
Lines down the page	Dirty heads/drum	Clean or replace cartridge
Garbled print	Wrong/corrupt driver	Reinstall driver
Paper jam	Misaligned/torn paper, worn rollers	Clear jam, check rollers
Faded prints	Low toner, misaligned head	Replace cartridge, adjust density
Paper not feeding	Dirty rollers, tray issues	Clean rollers, reload paper
Speckling	Toner leaks, dirty components	Replace toner, clean interior
Double/echo images	Fuser or drum issue	Replace fuser or drum
Staple/hole punch issues	Jammed cartridge, debris	Clear jam, clean unit
Tray not recognized	Misalignment, driver config	Reseat tray, check driver
Frozen print queue	Spooler crash or corrupt job	Restart spooler, clear jobs

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